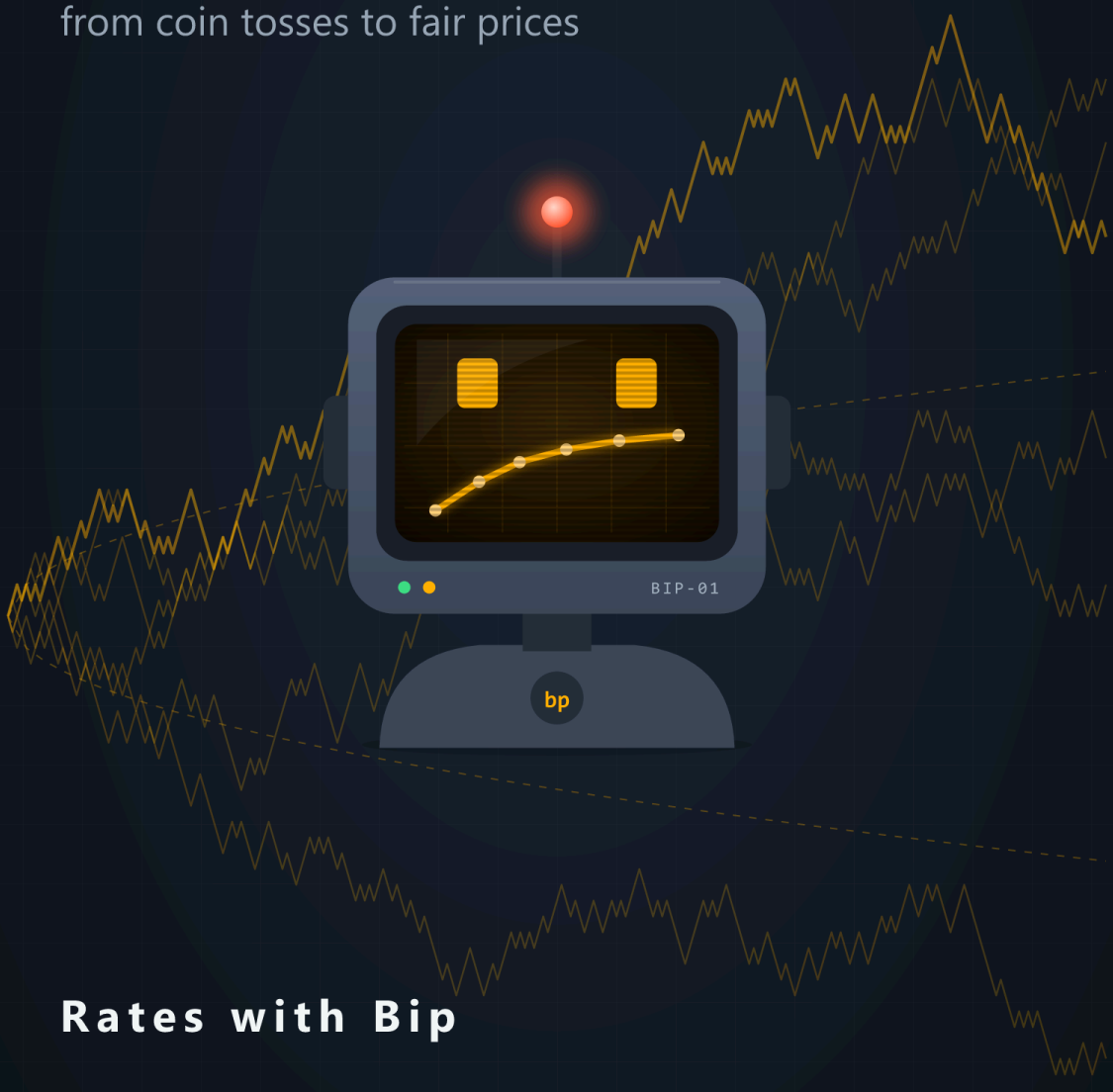
QUANT SERIES · PART 2 OF 5

Stochastic Calculus, Step by Step

Part 2: discrete models,
from coin tosses to fair prices



Rates with Bip

STOCHASTIC CALCULUS, STEP BY STEP · PART 2 OF 5

Stochastic Calculus, Step by Step

Part 2: Discrete models, with Python

Rates with Bip

Stochastic Calculus, Step by Step. Part 2: Discrete models, with Python.

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Part 2 works on one teaching tree: a share at $S_0 = 100$ that goes up 10% or down 10% every four months, three steps to the course's one-year horizon, with a real-world chance of 0.6 of going up, a risk-free rate $r = 3\%$ (continuously compounded) and a strike $K = 105$. Lessons 10 and 12 also use the course's volatility $\sigma = 20\%$. All made up for teaching; every random number is simulated with a fixed seed. There is no market data.

The exercises are at the end of each lesson. Their full solutions, and the answers to the self-test questions, are in the companion volume, Stochastic Calculus, Step by Step, Part 2: Solutions.

Every number in the text, tables and figures is computed by the companion Python notebooks (numpy only), one per lesson, which reproduce each result.

This book is for education only. It is not financial, investment, legal or tax advice.

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How to use this book

This is Part 2 of *Stochastic Calculus, Step by Step*, a course in five parts that goes from school calculus to the Black-Scholes formula, Feynman-Kac and the numerical methods banks use to price options. Part 1 laid the mathematical foundations. Part 2 builds the whole theory of option pricing in a world small enough to see completely: a share that moves up or down a few times. Every idea of the continuous-time theory in Parts 3 and 4 appears here first, in a form you can check by hand.

LESSON	TOPIC	USED LATER FOR
7	Information as σ-algebras, filtrations, probability measures	adapted processes, the Itô integral
8	Conditional expectation and its rules: the tower rule above all	martingales, every pricing formula
9	One step: no arbitrage, copying a call, the risk-neutral probability	the Black-Scholes argument
10	Many steps: backward induction, dynamic hedging, put-call parity	delta hedging, numerical methods
11	Martingales, the risk-neutral pricing formula, the fundamental theorems	risk-neutral pricing, Feynman-Kac
12	Changing the probability, state prices, American options	Girsanov's theorem, early exercise

Who this book is for

As in Part 1, this book is written for readers learning on their own, without a teacher to fill the gaps. Every new idea is **defined** before it is used, **explained** in words, tried on a small example, and then **derived** one line at a time, with the reason for each line. When a step uses an earlier lesson, the lesson is named.

You need Part 1's probability: expectation, variance, independence (Lesson 3) and the law of large numbers (Lesson 6). Lessons 10 and 12 compare the tree with the Black-Scholes value, which uses the normal distribution function N of Lesson 4; you can take those numbers on trust until Part 4 derives them.

One tree for the whole of Part 2

Part 1 used one set of numbers throughout. Part 2 adds one **teaching tree**, small enough to work by hand:

- the share is worth $S_0 = 100$ today;
- every four months it goes **up 10%** (we call this *heads*, H) or **down 10%** (*tails*, T), so after one step it is 110 or 90;
- three steps make the course's horizon of one year, so there are $2^3 = 8$ possible paths, such as HTH;
- the real-world chance of heads is $p = 0.6$;
- cash earns the course's risk-free rate $r = 3\%$, continuously compounded, so 1 in cash grows to $R = e^{0.03/3} \approx 1.01005$ in four months;
- options have the course's strike, $K = 105$.

Lessons 10 and 12 also build big trees from the course's volatility, $\sigma = 20\%$, to show that the tree's prices settle on the Black-Scholes values.

How each lesson is built

Every lesson follows the same steps as in Part 1: **why this lesson**, **what you need** from earlier, **the picture**, the derivations as numbered steps, **numbers and code**, **what to remember**, **common mistakes**, eight **exercises** (six by hand, two in code) and a three-question **self-test**.

How to read it

Read with a pencil, and draw the tree. Almost every result in Part 2 can be checked by writing the eight paths in a column and adding up. When a derivation says "Step 2", try it yourself before reading it. Run the notebook next to the text and change the numbers: make the share move 20% instead of 10%, or set $p = 0.9$, and watch which answers move and which don't.

Notebooks and solutions

Every number, table and figure is computed by a companion Python notebook, one per lesson, using numpy only. The notebooks are free at github.com/RatesWithBip/curriculum.

The solutions are **not** in this book: they are in the companion volume, *Part 2: Solutions*, with every derivation written out and every code solution shown with what it prints. Try each exercise before you look.

Reading alongside

Steven Shreve's *Stochastic Calculus for Finance I* (Chapters 1 to 4) covers the same ground in more depth, and Martin Baxter and Andrew Rennie's *Financial Calculus*, Chapter 2, is a short and lively tour of it. This book is written fresh from its own notebooks and uses none of their examples or exercises.

Free, and staying free

The videos, the notebooks and both books are free. If they help you, you can support the channel with a coffee at ko-fi.com/rateswithbip.

This book is for education only and is not financial advice.

LESSON 7 · VIDEO 7

Information and Measure

The coin-toss model that runs through Part 2; events, and information described as a partition into blocks; σ-algebras, what their three rules mean and why information obeys them; filtrations, information that grows with time; measurability, what you can compute from what you know; probability measures and their three rules; and why, with infinitely many tosses, probability has to be defined on a σ-algebra.

Why this lesson

An option pays according to what the share does. A trader decides what to do according to **what they know so far**. Part 1 never had to say what "what they know" means: it worked with one random number at a time. From now on we follow prices through time, and three questions keep coming back:

- At time n, which questions about the future can already be answered?
- Which quantities can be computed from the information available at time n? A hedge ratio must be one of them, or nobody could trade it.
- How do we give probabilities to statements about the future?

The mathematical answers are a **σ-algebra** (information), a **filtration** (information growing in time), **measurability** (computable from the information) and a **probability measure**. These words sound forbidding, but on a small tree each one is something you can draw. This lesson draws them.

Every later lesson uses this language. Conditional expectation (Lesson 8) is "the best guess given a σ-algebra"; a martingale (Lesson 11) is defined with a filtration; the Itô integral (Lesson 15) only makes sense because the integrand is measurable at the left end of each step.

What you need from earlier lessons

- **Outcomes, events, probability** (Lesson 3). An event is a statement about the outcome that is true or false; P gives each event a number between 0 and 1.

- **Random variables** (Lesson 3). A number that depends on the outcome.
- **Independence** (Lesson 3). For independent tosses, probabilities multiply.
- **The law of large numbers** (Lesson 6). The share of heads in many tosses settles at the probability of heads.
- **Sets.** A set is a collection of things, written in braces: $\{\mathrm{HHH}, \mathrm{HHT}\}$. $A \cup B$ (union) is everything in A or B; $A \cap B$ (intersection) is everything in both; A^c (complement) is everything not in A; $\emptyset$ is the empty set; $A \subseteq B$ means every element of A is in B.

The picture

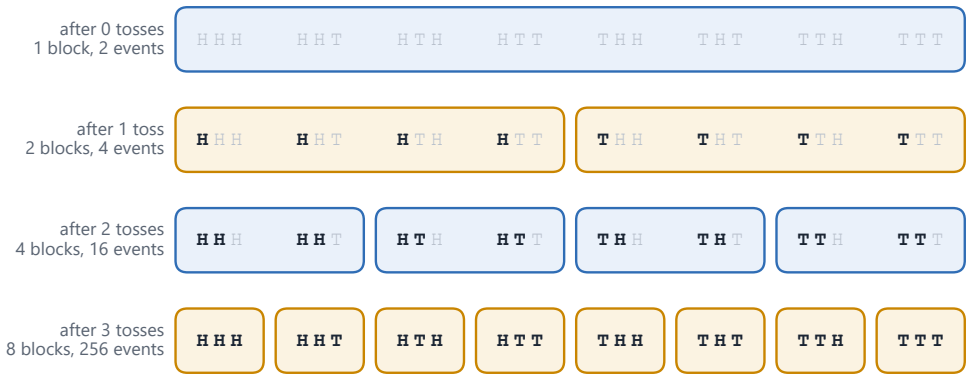


Figure 7.1. The eight possible worlds of three tosses, and what you know after 0, 1, 2 and 3 tosses. Tosses already seen are in ink; the rest are grey. Worlds in the same box cannot yet be told apart. Each new toss splits every box in two, and the number of questions you can answer squares.

Before any toss, all eight worlds are possible and you cannot tell them apart: one big box. After the first toss you know the first letter, so the worlds split into two boxes. After two tosses there are four boxes, and after three, eight: you know everything. **Information is the way the worlds are split into boxes.** The rest of this lesson turns that picture into definitions.

The coin-toss model

The share is worth $S_0 = 100$ today. Every four months a coin is tossed: heads (H) and the share rises 10%, tails (T) and it falls 10%. So with $u = 1.1$ (up) and $d = 0.9$ (down),

$$S_{n+1} = \begin{cases} u\,S_n & \text{if toss } n+1 \text{ is H}, \\ d\,S_n & \text{if toss } n+1 \text{ is T}. \end{cases}$$

After three tosses, a year has passed.

Worlds. A **world** (or outcome) ω is one complete sequence of three tosses, such as HTH. There are $2 \times 2 \times 2 = 8$ of them, and the set of all of them is

$$\Omega = \{\mathrm{HHH}, \mathrm{HHT}, \mathrm{HTH}, \mathrm{HTT}, \mathrm{THH}, \mathrm{THT}, \mathrm{TTH}, \mathrm{TTT}\}.$$

Each world fixes the whole price path. For example, in the world HTH the price goes $100 \to 110 \to 99 \to 108.9$. In general, if the first n tosses of ω contain h heads and $n - h$ tails,

$$S_n(\omega) = 100 \times 1.1^{\,h} \times 0.9^{\,n-h}.$$

Only the number of heads matters, not their order: HT and TH both give $100 \times 1.1 \times 0.9 = 99$.

Probabilities. The tosses are independent and each is heads with probability $p = 0.6$. By independence (Lesson 3), the probability of a world is the product over its tosses:

$$P(\omega) = 0.6^{\#\mathrm{H}} \times 0.4^{\#\mathrm{T}}.$$

So $P(\mathrm{HHH}) = 0.6^3 = 0.216$, $P(\mathrm{HTH}) = 0.6 \times 0.4 \times 0.6 = 0.144$ and $P(\mathrm{TTT}) = 0.4^3 = 0.064$. Here is every world, with its price path and probability:

The eight worlds of the teaching tree. The probabilities add to 1.

WORLD	S_1	S_2	S_3	P
HHH	110.00	121.00	133.10	0.216
HHT	110.00	121.00	108.90	0.144
HTH	110.00	99.00	108.90	0.144
HTT	110.00	99.00	89.10	0.096
THH	90.00	99.00	108.90	0.144
THT	90.00	99.00	89.10	0.096
TTH	90.00	81.00	89.10	0.096
TTT	90.00	81.00	72.90	0.064

The probabilities add up to 1, as they must. (Check: grouping by the number of heads, $0.216 + 3(0.144) + 3(0.096) + 0.064 = 1$.)

Events

Definition. An **event** is a set of worlds: the worlds in which some statement is true.

Examples:

- "The first toss is heads" is $\{\mathrm{HHH}, \mathrm{HHT}, \mathrm{HTH}, \mathrm{HTT}\}$.
- "The share ends above 105" is the set of worlds with $S_3 > 105$. From the table, S_3 is 133.1 with three heads, 108.9 with two, 89.1 with one and 72.9 with none, so this is "at least two heads": $\{\mathrm{HHH}, \mathrm{HHT}, \mathrm{HTH}, \mathrm{THH}\}$.
- Ω itself is the event "something happens"; the empty set $\emptyset$ is the event "nothing happens", which is never true.

With eight worlds, an event is any choice of which worlds are in it: each world is in or out, so there are $2^8 = 256$ events in all.

Statements combine, and so do events. "A or B" is the union $A \cup B$; "A and B" is the intersection $A \cap B$; "not A" is the complement A^c.

Information as a partition

Suppose two tosses have happened and they were H then T. What do you know? The world is HTH or HTT, and you cannot tell which. In general, after n tosses you know the first n letters of the world and nothing more.

Definition. The **time-n blocks** are the groups of worlds that agree on their first n tosses. Together they form a **partition** of Ω: every world is in exactly one block.

- After 0 tosses: one block, Ω.
- After 1 toss: two blocks, $\{\mathrm{H}{*}{*}\}$ and $\{\mathrm{T}{*}{*}\}$, where $*$ means "either".
- After 2 tosses: four blocks, $\{\mathrm{HH}{*}\}$, $\{\mathrm{HT}{*}\}$, $\{\mathrm{TH}{*}\}$, $\{\mathrm{TT}{*}\}$.
- After 3 tosses: eight blocks of one world each.

This is the picture at the top of the lesson. At time n you know **which block** you are in, and nothing that would separate worlds inside a block.

Which events are known at time n? An event A is *known at time n* if, at time n, you can always say whether A happened. That means every block lies entirely inside A or entirely outside it: if a block were split, some worlds in it would be in A and some not, and you could not tell which you were in. So:

> **An event is known at time n exactly when it is a union of time-n blocks.**

Example. Is "the share ends above 105" known after two tosses? The time-2 block $\{\mathrm{HH}*\}$ is inside it (both HHH and HHT end above 105) and $\{\mathrm{TT}*\}$ is outside it. But $\{\mathrm{HT}*\}$ is split: HTH ends at 108.9, above 105, and HTT at 89.1, below. So the event is **not** known at time 2; it becomes known only at time 3 in the worlds starting HT or TH, and already at time 2 in those starting HH or TT.

σ-algebras

The events known at time n form a collection with three properties that every sensible notion of "information" must have.

Definition. A collection $\mathcal{F}$ of events is a **σ-algebra** (say "sigma-algebra") on Ω if

1. $\Omega \in \mathcal{F}$;
2. if $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$;
3. if $A_1, A_2, \cdots \in \mathcal{F}$ (a finite or countably infinite list), then $A_1 \cup A_2 \cup \cdots \in \mathcal{F}$.

Read $A \in \mathcal{F}$ as "with the information $\mathcal{F}$, you can tell whether A happened". Then the rules say:

1. You can always tell whether *something* happened.
2. If you can tell whether A happened, you can tell whether it didn't.
3. If you can tell for each of $A_1, A_2, \ldots$, you can tell whether at least one of them happened.

Two more facts follow at once. $\emptyset = \Omega^c$ is in $\mathcal{F}$ (rules 1 and 2). And intersections are in $\mathcal{F}$: "A and B" is "not (not A or not B)", that is, $A \cap B = (A^c \cup B^c)^c$, which uses only rules 2 and 3.

(The σ in the name refers to rule 3 allowing countably infinite lists. On a finite Ω there are only finitely many events, so finite unions are enough.)

The σ-algebra generated by a partition

Claim. Take a partition of Ω into blocks $B_1, \ldots, B_b$, and let $\mathcal{F}$ be the collection of all unions of blocks (including the empty union, $\emptyset$). Then $\mathcal{F}$ is a σ-algebra, and it has exactly 2^b events.

Why it is a σ-algebra.

1. Ω is the union of all the blocks.
2. The complement of a union of some blocks is the union of the other blocks.
3. A union of unions of blocks is again a union of blocks.

Why 2^b events. An event in $\mathcal{F}$ is fixed by choosing, for each block, whether it is in or out: two choices per block, b blocks, so 2^b choices. Different choices give different events, because the blocks don't overlap.

The time-n σ-algebra $\mathcal{F}_n$ is the one generated by the time-n blocks: the collection of all events known at time n. Counting blocks:

AFTER	BLOCKS	EVENTS IN $\mathcal{F}_N$
0 tosses	1	2
1 toss	2	4
2 tosses	4	16
3 tosses	8	256

$\mathcal{F}_0 = \{\emptyset, \Omega\}$ is the **trivial σ-algebra**: before any toss you can only answer questions whose answer is certain. $\mathcal{F}_3$ contains all 256 events: after three tosses you know everything. The notebook builds each $\mathcal{F}_n$ as all unions of blocks and checks the three rules by brute force.

The three rules, checked by brute force.

```
Omega = frozenset(worlds)


def is_sigma_algebra(F):
    F = set(F)
```

A collection that is not a σ-algebra. $\{\emptyset, \Omega, \{\text{first toss H}\}\}$ breaks rule 2: it contains "the first toss is heads" but not "the first toss is tails". Information that could answer one but not the other makes no sense.

Filtrations

Information only grows: what you know after one toss, you still know after two. In terms of events, every event known at time n is also known at time $n+1$, because each time-n block is a union of two time-$(n+1)$ blocks. So

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \mathcal{F}_3.$$

Definition. A **filtration** is a sequence of σ-algebras $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \cdots$, one for each time, each containing the one before. It models information arriving over time and never being forgotten.

In the picture, the filtration is the stack of rows: each row's boxes are the previous row's boxes cut in two.

Measurability: what you can compute from what you know

Definition. A random variable X is **$\mathcal{F}_n$-measurable** (or *known at time n*) if its value can be computed from the time-n information. On the tree, that means X takes **one value on each time-n block**.

Equivalently, every event of the form $\{X \le c\}$ is in $\mathcal{F}_n$: you can tell whether X is at most c. (If X is constant on blocks, $\{X \le c\}$ is the union of the blocks where that constant is at most c; and if X took two values in one block, some $\{X \le c\}$ would split the block.)

Examples. Read each one off the table of worlds.

RANDOM VARIABLE	KNOWN AT TIME 0?	1?	2?	3?
S_1	no	yes	yes	yes
S_2	no	no	yes	yes
S_3	no	no	no	yes
$\max(S_0, S_1, S_2)$	no	no	yes	yes
$\mathbf{1}\{S_3 > 105\}$	no	no	no	yes

For instance S_2 is not $\mathcal{F}_1$-measurable: the time-1 block $\{\mathrm{H}{*}{*}\}$ contains HHH with $S_2 = 121$ and HTH with $S_2 = 99$. (Here $\mathbf{1}\{A\}$, the **indicator** of A, is 1 in the worlds of A and 0 elsewhere.)

Definition. A process $X_0, X_1, X_2, \dots$ is **adapted** to the filtration if each X_n is $\mathcal{F}_n$-measurable: its value at time n is known at time n.

The share price S_n is adapted. So is the running maximum $\max(S_0, \dots, S_n)$. A trading strategy must be adapted too: the number of shares you hold from time n to $n+1$ has to be decided at time n, with time-n information. A strategy that used S_{n+1} to decide what to hold before S_{n+1} is revealed would be trading on tomorrow's newspaper. This is the condition that makes the Itô integral of Part 3 work.

The information in a random variable

A random variable carries information of its own: knowing its value splits the worlds into blocks with the same value.

Definition. $\sigma(X)$, the **σ-algebra generated by** X, is the collection of events you can decide from the value of X alone: the unions of the blocks $\{X = x\}$.

Example: $\sigma(S_2)$. S_2 is 121, 99 or 81. The worlds with $S_2 = 99$ are HTH, HTT, THH and THT: four worlds, one block, because HT and TH lead to the same price. So $\sigma(S_2)$ has three blocks and $2^3 = 8$ events, while $\mathcal{F}_2$ has four blocks and 16 events. Watching only the price, you lose the order of the moves: "the first toss was heads" is in $\mathcal{F}_2$ but not in $\sigma(S_2)$.

Whether that lost information matters depends on the question. For the future of the share price it does not (Lesson 11 calls this the **Markov property**). For a payoff that depends on the path, such as the highest price so far, it does.

Probability measures

Definition. A **probability measure** on $(\Omega, \mathcal{F})$ is a function P that gives each event $A \in \mathcal{F}$ a number $P(A)$, such that

1. $P(A) \geq 0$ for every $A \in \mathcal{F}$;
2. $P(\Omega) = 1$;
3. for any finite or countably infinite list of **disjoint** events (no two can happen together) $A_1, A_2, \ldots$,

$$P(A_1 \cup A_2 \cup \cdots) = P(A_1) + P(A_2) + \cdots .$$

The triple $(\Omega, \mathcal{F}, P)$ is called a **probability space**. These are the rules of Lesson 3, now stated for exactly the events in $\mathcal{F}$.

Consequences. Everything else follows from the three rules (Exercise 7.5 proves these):

- $P(\emptyset) = 0$;
- $P(A^c) = 1 - P(A)$;
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ for any two events.

On a finite tree, weights are enough. Every event is a disjoint union of single worlds, so by rule 3, $P(A) = \sum_{\omega \in A} P(\omega)$. Give each world a weight $P(\omega) \geq 0$ with total 1, and P is fixed on all 256 events.

Worked example

$P(S_3 > 105)$. The event is "at least two heads": HHH, HHT, HTH, THH. Add their weights:

$$P(S_3 > 105) = 0.6^3 + 3 \times 0.6^2 \times 0.4 = 0.216 + 0.432 = 0.648.$$

$P(\text{first toss H})$. The worlds H$**$ have weights $0.216 + 0.144 + 0.144 + 0.096 = 0.6$, which is just p, as it should be.

Both, and either. "Ends above 105 and starts with H" is HHH, HHT, HTH: $0.216 + 0.144 + 0.144 = 0.504$. Then

$$P(S_3 > 105 \text{ or first H}) = 0.648 + 0.6 - 0.504 = 0.744.$$

Check by listing: the union is HHH, HHT, HTH, HTT, THH, with weights adding to the same number.

At least one head. By the complement rule, $1 - P(\text{TTT}) = 1 - 0.4^3 = 0.936$.

Simulation. The notebook tosses three coins 100,000 times. The share of runs with $S_3 > 105$ is 0.6504, close to 0.648 (Lesson 6: averages settle down).

Different measures on the same worlds. Nothing forces $p = 0.6$. With a fair coin, $p = 0.5$, every world has weight $1/8$ and $P(S_3 > 105) = 4/8 = 0.5$. The worlds, the events and the information are the same; only the weights change. Lesson 9 introduces a second set of weights, the **risk-neutral probability**, and Lesson 12 studies how to move from one set of weights to another.

Why σ-algebras at all: infinitely many tosses

On the tree, every set of worlds is an event and we could have skipped σ-algebras for probability (though not for information). They become essential when time goes on for ever, or becomes continuous, as in Parts 3 and 4.

Toss the coin for ever. A world is now an infinite sequence such as HTTHTHH..., and there are uncountably many of them.

- **Each single world has probability 0.** The world "heads every time" is in the event "the first n tosses are heads" for every n, so its probability is at most 0.6^n: $0.6^{10} = 0.0060$, $0.6^{50} = 8.1\text{e-}12$, and so on to 0. It is possible, but has probability 0. So we can no longer get $P(A)$ by adding up world weights: they are all zero.
- **Some events still have clear probabilities.** "The first toss is heads" has probability 0.6. "The share of heads in the first n tosses tends to 0.6" has probability 1 (the law of large numbers): the notebook's simulated share of heads is 0.52 after 100 tosses, 0.6006 after 10,000 and 0.6024 after 100,000.
- **Not every set can be given a probability.** It is a theorem (too long for this course) that there is no way to give *every* subset of the infinite-toss Ω a probability that obeys the three rules and treats the tosses as fair and independent. So probability

has to be defined on a smaller collection of events, and the right collection is a σ-algebra: the one generated by the events "toss n is heads".

This is why every probability space has three parts, $(\Omega, \mathcal{F}, P)$. In Part 3, Ω is the set of all continuous paths of Brownian motion, $\mathcal{F}_t$ is the information in the path up to time t, and everything in this lesson carries over word for word.

What to remember

- A **world** ω is one complete path of tosses; Ω is the set of all worlds. An **event** is a set of worlds.
- **Information** at time n is a partition of Ω into blocks of worlds you cannot yet tell apart. An event is **known at time** n when it is a union of time-n blocks.
- A **σ-algebra** contains Ω and is closed under complements and (countable) unions. A partition into b blocks generates a σ-algebra with 2^b events.
- A **filtration** $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \cdots$ is information growing over time.
- X is **$\mathcal{F}_n$-measurable** when its value is known at time n: one value per block. A process is **adapted** when each X_n is known at time n. Trading strategies must be adapted.
- $\sigma(X)$ is the information in X alone; $\sigma(S_2)$ is coarser than $\mathcal{F}_2$ because HT and TH give the same price.
- A **probability measure** is non-negative, gives Ω probability 1 and adds over disjoint events. On a finite tree it is fixed by world weights.

Common mistakes

- **"An event is an outcome."** An event is a *set* of outcomes. "The share ends above 105" contains four worlds.
- **"$\mathcal{F}_n$ is a set of worlds."** It is a set of *events* (sets of worlds): $\mathcal{F}_1$ has four elements, each of them a set of worlds.
- **"Known at time 2 means it has happened by time 2."** It means you can *tell* by time 2 whether it happened. "The first toss is tails" is known at time 2 (and at time 1).
- **Counting blocks instead of events.** After two tosses there are 4 blocks but $2^4 = 16$ events.
- **"$\sigma(S_2) = \mathcal{F}_2$."** The price forgets the order of the moves; $\sigma(S_2)$ has 8 events, $\mathcal{F}_2$ has 16.
- **"Probability 0 means impossible."** With infinitely many tosses, "heads every time" can happen but has probability 0.

Exercises

Solutions are in *Part 2: Solutions*.

7.1 List the four events in $\mathcal{F}_1$ and check the three rules of a σ-algebra for them.

7.2 Show that a partition of Ω into b blocks generates exactly 2^b events. How many events are in $\mathcal{F}_n$ for the n-toss tree? How many digits does the number of events in $\mathcal{F}_{10}$ have?

7.3 Let $A = \{S_3 > 105\}$. Show that $A \notin \mathcal{F}_2$, and find, world by world, the first time at which you know whether A happened.

7.4 Find the blocks of $\sigma(S_2)$. Show that $\sigma(S_2) \subsetneq \mathcal{F}_2$ (contained in, but not equal to) by naming an event in $\mathcal{F}_2$ that is not in $\sigma(S_2)$. What do you lose by watching only the price?

7.5 From the three rules of a probability measure, prove that $P(\emptyset) = 0$, $P(A^c) = 1 - P(A)$ and $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. Then compute $P(S_3 > 105)$ and $P(S_3 > 105 \text{ or the first toss is H})$ on the teaching tree.

7.6 Let τ be the time of the first tail ($\tau = 4$ if there is none in three tosses). Show that $\{\tau \le n\} \in \mathcal{F}_n$ for each n, but that $\{\tau = 1\} \cup \{\tau = 3\}$ is not in $\mathcal{F}_2$. Compute $P(\tau = n)$ for $n = 1, 2, 3, 4$.

7.7 *(Code)* For the four-toss tree, build $\mathcal{F}_2$ as all unions of blocks and check by brute force that it is a σ-algebra with $2^4 = 16$ events.

7.8 *(Code)* Simulate 200,000 years of three tosses with $p = 0.6$. Estimate the probability of each time-2 block and of $\{S_3 > 105\}$, and compare with the exact values.

Self-test

1. After one toss, how many events can you decide?
2. Is S_3 known at time 2? Explain with one block.
3. Which tells you more: knowing S_2, or knowing the first two tosses?

Companion notebook: `s5e07_information_and_measure.ipynb`. Every number in this chapter is computed by it.

LESSON 8 · VIDEO 8

Conditional Expectation

The best estimate of a future quantity given what you know now; why that estimate is itself a random variable; the defining property (partial averages) that makes it unique; the four rules (linearity, taking out what is known, independence and the tower rule), each proved on the tree; averaging a payoff layer by layer; why it is the best prediction; and conditioning on the price alone.

Why this lesson

Lesson 3 averaged a random variable over all possible worlds: $E[X]$, one number. But a trader at time 1 has already seen the first toss. Averaging over worlds that are no longer possible would be silly. They want the average over the worlds that are **still possible given what they know**.

That is the **conditional expectation** $E[X \mid \mathcal{F}_n]$: the best estimate of X with the information available at time n. It is the single most important tool in this course:

- every price is a conditional expectation, $V_n = \tilde{E}[\text{payoff} \mid \mathcal{F}_n]$ discounted (Lesson 11);
- a martingale is defined by one (Lesson 11);
- backward induction on a tree (Lessons 9 and 10) is the tower rule of this lesson, applied step by step;
- in Part 4, Feynman-Kac turns a conditional expectation into a partial differential equation.

This lesson defines it on the teaching tree, where it is an ordinary weighted average, proves its rules there, and states the definition in the form that still works when the tree is replaced by Brownian motion.

What you need from earlier lessons

- **Expectation** (Lesson 3). $E[X] = \sum_\omega X(\omega)P(\omega)$ on a finite set of worlds.

- **Conditional probability** (Lesson 3). $P(A \mid B) = P(A \cap B)/P(B)$ when $P(B) > 0$: the share of B's probability that also lies in A.
- **Independence** (Lesson 3). For independent X and Y, $E[XY] = E[X]E[Y]$.
- **Blocks, σ-algebras and measurability** (Lesson 7). At time n you know which time-n block you are in. $\mathcal{F}_n$ is the σ-algebra of unions of those blocks. Y is $\mathcal{F}_n$-measurable when it takes one value on each block. $\mathbf{1}_A$ is 1 on A and 0 off it.
- **The teaching tree** (Lesson 7). $S_0 = 100$; each step multiplies the price by $u = 1.1$ (heads) or $d = 0.9$ (tails); $P(\mathrm{H}) = p = 0.6$; three steps.

The picture

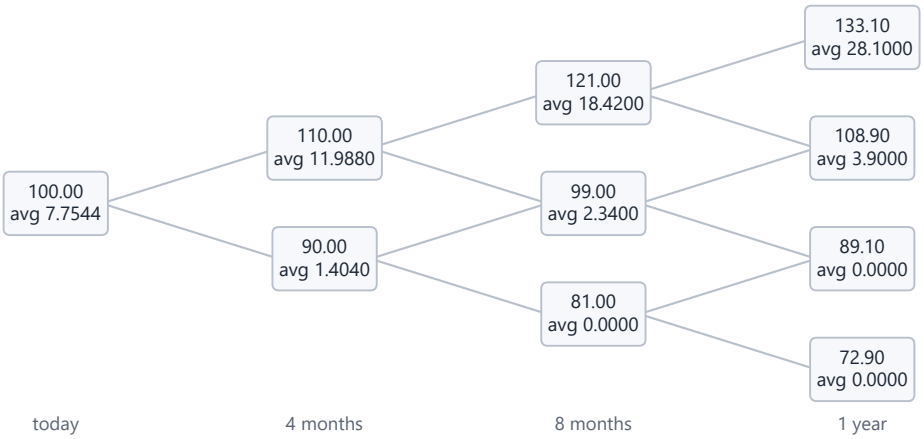


Figure 8.1. The expected call payoff E[max(S₃ − 105, 0)], computed one layer at a time. Each node shows the share price and the average payoff seen from that node: the payoffs at one year are averaged over the last toss (weights 0.6 and 0.4) to give the values at eight months, those are averaged over the second toss, and so on back to today.

At expiry the call pays $\max(S_3 - 105, 0)$: 28.1, 3.9, 0 or 0. One step earlier, at the node where the price is 121, the call will pay 28.1 or 3.9 with probabilities 0.6 and 0.4, so its average payoff seen from there is $0.6 \times 28.1 + 0.4 \times 3.9 = 18.42$. Repeating this back to today gives 7.7544, the same as averaging the eight paths at once. Each number on this tree is a **conditional expectation**. (They are real-world averages, not prices; Lesson 9 shows how prices differ.)

Averaging over a block

The idea

At time n you know which block you are in. Within that block, worlds still have their probabilities, but the worlds outside the block have been ruled out. So:

> **$E[X \mid \mathcal{F}_n]$, in world ω, is the average of X over the time-n block that contains ω, with each world weighted by its probability.**

The formula

Write $B(\omega)$ for the time-n block containing ω. The weights of the worlds inside B add to $P(B)$, not 1, so divide by $P(B)$ to make them add to 1:

$$E[X \mid \mathcal{F}_n](\omega) = \frac{\sum_{v \in B(\omega)} X(v)\, P(v)}{P\big(B(\omega)\big)}.$$

The weight $P(v)/P(B)$ given to world v is exactly the conditional probability $P(\{v\} \mid B)$ of Lesson 3. So this is the ordinary expectation, but with the probabilities **updated** to what you know.

Worked example: the share price in a year, seen after one toss

Take $X = S_3$ and $n = 1$. There are two time-1 blocks.

Block H, the four worlds starting with heads. $P(\mathrm{H}{*}{*}) = 0.6$. The worlds, their prices S_3 and weights:

WORLD	S_3	$P(\Omega)$	$P(\Omega)/0.6$
HHH	133.1	0.216	0.36
HHT	108.9	0.144	0.24
HTH	108.9	0.144	0.24
HTT	89.1	0.096	0.16

The updated weights in the last column add to 1. Averaging, in block H,

$$\begin{aligned} E[S_3 \mid \mathcal{F}_1] &= 133.1(0.36) + 108.9(0.48) + 89.1(0.16) \\ &= 47.916 + 52.272 + 14.256 = 114.444. \end{aligned}$$

Block T. The same steps with the four worlds starting with tails give 93.636.

A shortcut for the share price

There is a pattern. Seen from any node, the next price is uS_n or dS_n with probabilities p and $1-p$, so its average is

$$E[S_{n+1} \mid \mathcal{F}_n] = p\,uS_n + (1-p)\,dS_n = \big(pu + (1-p)d\big)\,S_n = 1.02\,S_n.$$

The factor $m = pu + (1-p)d = 0.6(1.1) + 0.4(0.9) = 1.02$ is the average growth per step. Each further step multiplies by another 1.02 (the tower rule, below, makes this precise), so

$$E[S_3 \mid \mathcal{F}_n] = 1.02^{\,3-n} S_n.$$

Check. In block H, $S_1 = 110$ and $1.02^2 \times 110 = 1.0404 \times 110 = 114.444$. In block T, $1.0404 \times 90 = 93.636$. The notebook computes $E[S_3 \mid \mathcal{F}_n]$ by averaging over blocks for every world and every n, and checks the shortcut:

$E[S_3 \mid F_n]$ in each world. Worlds in the same time-n block share a value; at n = 3 it is S_3 itself.

WORLD	TIME 0	TIME 1	TIME 2	TIME 3
HHH	106.121	114.444	123.420	133.100
HHT	106.121	114.444	123.420	108.900
HTH	106.121	114.444	100.980	108.900
HTT	106.121	114.444	100.980	89.100
THH	106.121	93.636	100.980	108.900
THT	106.121	93.636	100.980	89.100
TTH	106.121	93.636	82.620	89.100
TTT	106.121	93.636	82.620	72.900

Read the table column by column. At $n = 0$ there is one block, so one value: the plain expectation $E[S_3] = 106.1208$. At $n = 1$ there are two values, at $n = 2$ four, and at $n = 3$, when everything is known, the "estimate" is just S_3.

Why it is a random variable

$E[X]$ is a number. $E[X \mid \mathcal{F}_1]$ is not: **before** the first toss you don't know which block you will be in, so you don't know which average will apply. It will be 114.444 with probability 0.6 and 93.636 with probability 0.4. That makes it a random variable, with three properties worth saying out loud:

1. **It is known at time** n ($\mathcal{F}_n$-measurable): it takes one value on each time-n block, by construction.

2. **It has its own distribution.** $E[S_3 \mid \mathcal{F}_1]$ takes two values with probabilities 0.6 and 0.4.
3. **Its average is the plain average.** $0.6 \times 114.444 + 0.4 \times 93.636 = 106.1208 = E[S_3]$. Averaging the block averages, with the block probabilities as weights, gives back the overall average. (This is the tower rule in its simplest form.)

Notation. People also write $E_n[X]$ or $E[X \mid S_1]$ (when $\mathcal{F}_1$ is the information in S_1). When we want its value in a particular block, we write e.g. $E[S_3 \mid \mathcal{F}_1](\mathrm{H})$.

The defining property: partial averages

The block-average formula works on a tree. For Brownian motion every single path has probability 0, and dividing by $P(B)$ breaks down. So we need a definition that does not divide by anything. It comes from a property the block average has.

The property. Let $Y = E[X \mid \mathcal{F}_n]$ and take any event $A \in \mathcal{F}_n$. Then

$$E[Y\,\mathbf{1}_A] = E[X\,\mathbf{1}_A].$$

In words: **over any event you can decide at time n, the estimate and the real thing have the same total**. That is why it's called the **partial-averaging** property: it holds not just for the whole average but for partial averages over events in $\mathcal{F}_n$.

Proof on the tree.

Step 1. One block. Let B be a single time-n block. On B, Y is the constant $y_B = \sum_{v \in B} X(v)P(v)/P(B)$. So

$$E[Y\,\mathbf{1}_B] = \sum_{\omega \in B} y_B\,P(\omega) = y_B\,P(B) = \sum_{v \in B} X(v)P(v) = E[X\,\mathbf{1}_B].$$

Step 2. Any event in $\mathcal{F}_n$. Such an event A is a union of blocks (Lesson 7). The blocks don't overlap, so $\mathbf{1}_A$ is the sum of the blocks' indicators, and by linearity both sides add up block by block. Step 1 gives each block's equality, so the sums are equal.

The general definition. $E[X \mid \mathcal{G}]$ is the random variable Y such that

1. Y is $\mathcal{G}$-measurable, and
2. $E[Y\,\mathbf{1}_A] = E[X\,\mathbf{1}_A]$ for every $A \in \mathcal{G}$.

Why only one Y fits. Suppose Y and Y' both satisfy 1 and 2. On the tree, take A to be one block B: both are constant on B, and both give $E[\,\cdot\,\mathbf{1}_B] = E[X\mathbf{1}_B]$, so $y_B P(B) = y'_B P(B)$, and if $P(B) > 0$ the constants agree. (In general one shows $P(Y \neq Y') = 0$.)

So the two conditions *define* the conditional expectation, and this is the definition used from Part 3 on.

Numbers. For the call payoff $X = \max(S_3 - 105, 0)$ and all 16 events of $\mathcal{F}_2$, the notebook computes both sides; the largest gap is 8.9e-16, rounding noise. For $A =$ "first toss heads" both sides are 7.1928.

Four rules

These rules let you compute conditional expectations without going back to blocks. Each is proved here on the tree; all of them hold in general, and Part 3 uses them constantly. Let $\mathcal{G}$ be the information at some time (a σ-algebra generated by blocks).

Rule 1. Linearity

$$E[aX + bY \mid \mathcal{G}] = a\,E[X \mid \mathcal{G}] + b\,E[Y \mid \mathcal{G}].$$

Why. In each block, the left side is a weighted average of $aX + bY$, and a weighted average of a sum is the sum of the weighted averages (Lesson 3).

Rule 2. Taking out what is known

If Y is $\mathcal{G}$-measurable (known),

$$E[YX \mid \mathcal{G}] = Y\,E[X \mid \mathcal{G}].$$

Why. In each block Y is a constant, y_B, and a constant factor comes out of a weighted average:

$$\frac{\sum_{v \in B} y_B X(v) P(v)}{P(B)} = y_B \, \frac{\sum_{v \in B} X(v) P(v)}{P(B)}.$$

Meaning. Once something is known, it behaves like a number. **Example:** $E[S_1 S_3 \mid \mathcal{F}_1] = S_1\,E[S_3 \mid \mathcal{F}_1] = 1.0404\,S_1^2$.

A special case: if X itself is $\mathcal{G}$-measurable, $E[X \mid \mathcal{G}] = X$ (take the "X" in the rule to be 1). If you already know it, your best estimate is the thing itself.

Rule 3. Independence

If X is independent of the information $\mathcal{G}$,

$$E[X \mid \mathcal{G}] = E[X].$$

Why. Independence means that within each block, the distribution of X is the same as overall: $P(X = x \mid B) = P(X = x)$. So the block average equals the overall average, in every block.

Meaning. Information that has nothing to do with X does not change your estimate. **Example:** the third step's growth factor $G = S_3/S_2$ is u or d according to the third toss, which is independent of the first two. So $E[G \mid \mathcal{F}_2] = E[G] = 0.6(1.1) + 0.4(0.9) = 1.02$ in every block. Combined with Rule 2 (since S_2 is known at time 2):

$$E[S_3 \mid \mathcal{F}_2] = E[S_2\, G \mid \mathcal{F}_2] = S_2\, E[G \mid \mathcal{F}_2] = 1.02\, S_2.$$

That is the shortcut, now proved.

Rule 4. The tower rule

If $\mathcal{H} \subseteq \mathcal{G}$ (the information $\mathcal{H}$ is less than $\mathcal{G}$, for example $\mathcal{F}_1 \subseteq \mathcal{F}_2$),

$$E\big[E[X \mid \mathcal{G}] \mid \mathcal{H}\big] = E[X \mid \mathcal{H}].$$

In words: **estimating your future estimate, with less information, gives your current estimate.** The best guess today of what you'll think tomorrow is what you think today.

Why, on the tree. Each $\mathcal{H}$-block C is a union of $\mathcal{G}$-blocks $B_1, \ldots, B_j$ (finer information cuts each block into smaller ones). Write $Y = E[X \mid \mathcal{G}]$, equal to y_{B_i} on B_i. Averaging Y over C:

$$\frac{\sum_i y_{B_i} P(B_i)}{P(C)} = \frac{\sum_i \sum_{v \in B_i} X(v)P(v)}{P(C)} = \frac{\sum_{v \in C} X(v)P(v)}{P(C)},$$

using $y_{B_i} P(B_i) = \sum_{v \in B_i} X(v)P(v)$ from the definition. The right side is $E[X \mid \mathcal{H}]$ on C.

Special case. With $\mathcal{H} = \mathcal{F}_0$ (no information), $E[\,\cdot \mid \mathcal{F}_0]$ is the plain expectation, so

$$E\big[E[X \mid \mathcal{G}]\big] = E[X].$$

Worked check at block H. From the table, $E[S_3 \mid \mathcal{F}_2]$ is 123.42 on HH and 100.98 on HT. Within block H, the second toss is heads with probability 0.6, so

$$0.6 \times 123.42 + 0.4 \times 100.98 = 114.444 = E[S_3 \mid \mathcal{F}_1](\mathrm{H}).$$

The notebook checks all four rules in every world; the largest gaps are rounding noise.

Averaging layer by layer

The tower rule turns one big average into a sequence of small ones, which is exactly the picture at the top of the lesson. For a payoff X paid at time 3:

$$E[X] = E\Big[E\big[E[X \mid \mathcal{F}_2] \mid \mathcal{F}_1\big]\Big].$$

Work from the inside out:

Step 1. $E[X \mid \mathcal{F}_2]$: at each time-2 node, average the payoff over the last toss.

Step 2. $E[\,\cdot \mid \mathcal{F}_1]$: at each time-1 node, average the step-1 results over the second toss.

Step 3. Average those over the first toss.

For $X = \max(S_3 - 105, 0)$:

- At 8 months: node 121 gives $0.6(28.1) + 0.4(3.9) = 18.42$; node 99 gives $0.6(3.9) + 0.4(0) = 2.34$; node 81 gives 0.
- At 4 months: node 110 gives $0.6(18.42) + 0.4(2.34) = 11.988$; node 90 gives $0.6(2.34) + 0.4(0) = 1.404$.
- Today: $0.6(11.988) + 0.4(1.404) = 7.7544$.

Averaging the eight paths directly gives 7.7544, the same. Why do the nodes work, when worlds HT and TH were different blocks? Because the future payoff depends only on the current price, not on how it got there, so the two blocks have the same average. This is the **Markov property**, and it is what keeps trees small: three steps need 10 nodes, not 15 blocks. Lesson 11 returns to it.

Lessons 9 and 10 use the same backward averaging to *price* options. Two things change: the weights 0.6 and 0.4 are replaced by the risk-neutral probabilities, and each step is discounted.

The best prediction

Why call $E[X \mid \mathcal{F}_n]$ the *best* estimate? Because among everything you could compute at time n, it makes the smallest **mean squared error**.

Claim. If Z is any $\mathcal{F}_n$-measurable random variable, then

$$E\big[(X - Z)^2\big] \geq E\big[(X - E[X \mid \mathcal{F}_n])^2\big].$$

Why. Write $Y = E[X \mid \mathcal{F}_n]$ and split $X - Z = (X - Y) + (Y - Z)$. Squaring and averaging,

$$E[(X - Z)^2] = E[(X - Y)^2] + 2E\big[(X - Y)(Y - Z)\big] + E[(Y - Z)^2].$$

The middle term is 0. Indeed $Y - Z$ is known at time n, so by the tower rule and Rule 2,

$$E\big[(X-Y)(Y-Z)\big] = E\Big[(Y-Z)\,E[X-Y \mid \mathcal{F}_n]\Big] = E\big[(Y-Z)(Y-Y)\big] = 0.$$

So $E[(X-Z)^2] = E[(X-Y)^2] + E[(Y-Z)^2]$, which is smallest when $Z = Y$.

Numbers. Predict S_3 at time 1 with guesses of the form $c\,S_1$. The notebook tries a fine grid of c; the error is smallest at $c = 1.0404$, which is $1.02^2 = 1.0404$. Its mean squared error is 210.71, against 227.85 for "it stays where it is" ($c = 1$) and 314.63 for ignoring the first toss altogether (guessing $E[S_3]$).

Conditioning on the price alone

Often the information is "the value of some random variable", such as the price S_2. Then $E[X \mid S_2]$ means $E[X \mid \sigma(S_2)]$: average over the worlds with the same value of S_2.

Lesson 7 showed that $\sigma(S_2)$ is coarser than $\mathcal{F}_2$: HT and TH give the same price, 99, so they fall in one block. For $X = S_3$ this makes no difference. The future of the price depends only on the current price, so both averages are $1.02\,S_2$; the notebook's largest gap is 0.0. For a payoff that depends on the path, it can matter: Exercise 8.8 finds one.

What to remember

- $E[X \mid \mathcal{F}_n]$ is the probability-weighted average of X over the time-n block you are in: the best estimate of X with time-n information.
- It is a **random variable**, known at time n, with one value per block; at $n = 0$ it is $E[X]$, and once X is known it is X.
- **Definition** (works everywhere): $Y = E[X \mid \mathcal{G}]$ is $\mathcal{G}$-measurable and $E[Y\mathbf{1}_A] = E[X\mathbf{1}_A]$ for all $A \in \mathcal{G}$.
- **Rules:** linearity; taking out what is known, $E[YX \mid \mathcal{G}] = YE[X \mid \mathcal{G}]$; independence, $E[X \mid \mathcal{G}] = E[X]$; tower, $E[E[X \mid \mathcal{G}] \mid \mathcal{H}] = E[X \mid \mathcal{H}]$ for $\mathcal{H} \subseteq \mathcal{G}$, and $E[E[X \mid \mathcal{G}]] = E[X]$.
- On the tree: $E[S_{n+1} \mid \mathcal{F}_n] = (pu + (1-p)d)S_n$.
- Backward induction on a tree is the tower rule applied one step at a time.
- $E[X \mid \mathcal{F}_n]$ minimises the mean squared error among all time-n guesses.

Common mistakes

- "$E[X \mid \mathcal{F}_1]$ **is a number.**" It is a random variable: one number per time-1 block.

- **Forgetting to renormalise.** Inside a block the weights are $P(\omega)/P(B)$, not $P(\omega)$. Without the division, the "average" of a constant would not be that constant.
- **Taking out something that isn't known.** $E[S_2 S_3 \mid \mathcal{F}_1] \neq S_2\, E[S_3 \mid \mathcal{F}_1]$: S_2 is not known at time 1.
- **The tower rule the wrong way round.** $E[E[X \mid \mathcal{G}] \mid \mathcal{H}] = E[X \mid \mathcal{H}]$ needs $\mathcal{H} \subseteq \mathcal{G}$: the *outer* conditioning has the *less* information.
- **"Independent of $\mathcal{G}$" vs "known in $\mathcal{G}$".** Rule 3 is for X that $\mathcal{G}$ says nothing about; Rule 2 for Y that $\mathcal{G}$ determines. Most random variables are neither.
- **Confusing the average payoff with the price.** $E[\max(S_3 - 105, 0)] = 7.7544$ uses real-world weights and no discounting. It is not what the option costs (Lessons 9 to 11).

Exercises

Solutions are in *Part 2: Solutions*.

8.1 Compute $E[S_2 \mid \mathcal{F}_1]$ on each time-1 block directly from the definition (average over the block), and check that it equals $1.02\, S_1$.

8.2 Show that $E[S_{n+1} \mid \mathcal{F}_n] = \big(pu + (1-p)d\big) S_n$ for any n. For which p is the price expected to stay where it is?

8.3 Use *taking out what is known* to find $E[S_1 S_3 \mid \mathcal{F}_1]$, then use the tower rule to find $E[S_1 S_3]$.

8.4 Let $G = S_3/S_2$, the third step's growth factor. Show that G is independent of $\mathcal{F}_2$ and compute $E[G \mid \mathcal{F}_2]$. Why is $E[S_3 \mid \mathcal{F}_2] = S_2\, E[G]$?

8.5 Verify the tower rule by hand at the time-1 block H: compute $E[S_3 \mid \mathcal{F}_2]$ on the blocks HH and HT, average them, and compare with $E[S_3 \mid \mathcal{F}_1]$ on H.

8.6 For $X = \max(S_3 - 105, 0)$ and $A = \{\text{first toss H}\}$, check the partial-averaging property $E\big[E[X \mid \mathcal{F}_1]\mathbf{1}_A\big] = E[X\mathbf{1}_A]$ by computing both sides.

8.7 *(Code)* Write a function `cond_exp(X, n)` for the three-toss tree. Draw a random payoff X (eight random numbers, one per world) and check the tower rule $E\big[E[X \mid \mathcal{F}_2] \mid \mathcal{F}_1\big] = E[X \mid \mathcal{F}_1]$.

8.8 *(Code)* Let $M_3 = \max(S_0, S_1, S_2, S_3)$. Compute $E[M_3 \mid \mathcal{F}_2]$ and $E[M_3 \mid S_2]$ in every world. Where do they differ, and why?

Self-test

1. Is $E[S_3 \mid \mathcal{F}_1]$ a number or a random variable? What values can it take?
2. What is $E\big[E[X \mid \mathcal{F}_2]\big]$?
3. If Y is known at time 1, what is $E[Y X \mid \mathcal{F}_1]$?

Companion notebook: `s5e08_conditional_expectation.ipynb`. Every number in this chapter is computed by it.

LESSON 9 · VIDEO 9

The One-Period Binomial Model

One step of the teaching tree: the share, cash and a call; what an arbitrage is and why a market must not allow one; the condition $d < R < u$; copying the call with Δ shares and borrowed cash; how solving the copying equations produces the risk-neutral probability $\tilde{p}$; why the real probability of the share going up never enters the price; and how to trade against a wrong price.

Why this lesson

Start with a puzzle. In one step the share goes from 100 to 110 or 90, and a call with strike 105 pays 5 or nothing. The share goes up with probability 0.6. What is the call worth today?

The natural answer is "the average payoff, discounted": 0.6×5, divided by the growth of cash. That answer is **wrong**, and this lesson shows why. The right answer comes from a different idea: build a portfolio of shares and cash that pays exactly what the call pays, in both outcomes. Then the call must cost what the portfolio costs, or someone could make money for nothing.

Everything in the rest of the course grows from this one step:

- the **hedge ratio** Δ becomes the delta of the Black-Scholes model (Lesson 21);
- the **risk-neutral probability** $\tilde{p}$ becomes the risk-neutral measure of Lessons 11, 25 and 26;
- the fact that the real probability **disappears** becomes the fact that the share's expected return μ disappears from the Black-Scholes equation (Lesson 19).

What you need from earlier lessons

- **Expectation** (Lesson 3). A weighted average of the possible values.
- **Continuous compounding** (Lesson 1). Cash earning the rate r for time t grows by the factor e^{rt}.

- **Solving two linear equations in two unknowns.** Subtract one from the other to remove an unknown.
- **The teaching tree** (Lesson 7), first step only: $S_0 = 100$, then $S_0 u = 110$ (heads) or $S_0 d = 90$ (tails).

The picture

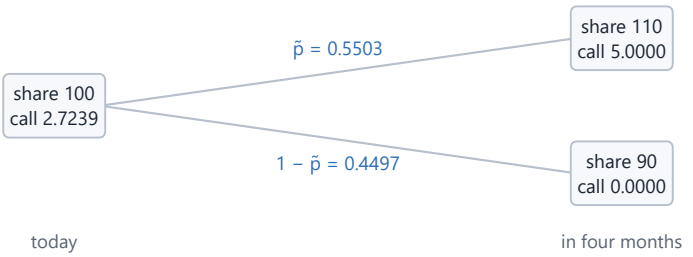


Figure 9.1. One step of the teaching tree: the share and a call with strike 105 that expires after four months. The call's price today, 2.7239, is the cost of a portfolio that pays exactly what the call pays. Writing that cost as a discounted average defines the risk-neutral probabilities on the branches.

Notice what is *not* on the picture: the real probability of heads, 0.6. It plays no part in the price.

The market in one step

Three things can be bought or sold today and are worth something in four months.

	TODAY	AFTER HEADS	AFTER TAILS
Share	100	110	90
Cash (1 deposited)	1	R	R
Call, strike 105	?	5	0

Cash. The risk-free rate is $r = 3\%$ a year, continuously compounded, so 1 deposited for four months ($t = 1/3$) grows to

$$R = e^{0.03/3} = e^{0.01} = 1.010050,$$

a gain of 1.0050%. Borrowing works the same way in reverse: a negative cash holding is a loan, and it grows by R too.

Assumptions. We can buy or sell any amount, including fractions of a share and short sales (selling a share you don't own and buying it back later); there are no trading costs; and borrowing and lending happen at the same rate. These are the standard assumptions of the theory. Real markets bend them, but the prices they lead to are the starting point everyone uses.

No arbitrage

Definition. An **arbitrage** is a trade that

1. costs nothing today (or pays you today),
2. can never lose money, and
3. makes money with positive probability (or for certain).

A free lunch. If one existed, everyone would do it, in unlimited size, and prices would move until it disappeared. So we look for prices under which **no arbitrage exists**. This single principle replaces any opinion about where the share is going.

The condition $d < R < u$

Even before the call, the share and cash alone could allow an arbitrage.

If cash always beats the share ($R \geq u$): sell one share short for 100 and deposit the 100. After four months you have $100R$ in cash and must buy back one share for 110 or 90. You are left with $100R - 110 \geq 0$ or $100R - 90 > 0$: never a loss, sometimes a gain. With, say, $R = 1.12$, that is a profit of 2 or 22, for nothing.

If the share always beats cash ($d \geq R$): borrow 100 and buy the share. You end with $110 - 100R > 0$ or $90 - 100R \geq 0$.

So no arbitrage requires

$$d < R < u.$$

Our market has $0.9 < 1.0101 < 1.1$. The share is risky: it might do better than cash or worse. (Later in this lesson we see that this condition is also *enough*: with it, there is no arbitrage at all.)

Copying the call

The idea

We can't trade the call yet, because we don't know its price. But we can try to **build it** from things we can trade: hold Δ shares (the Greek letter delta) and B in cash. If this portfolio pays exactly the call's payoff in both outcomes, it is a **copy** (a **replicating portfolio**) of the call.

The equations

After four months the portfolio is worth $\Delta \times (\text{share price}) + B \times R$. We want:

$$\begin{aligned} \text{after heads:} \quad & \Delta\, S_0 u + BR = V_u, \\ \text{after tails:} \quad & \Delta\, S_0 d + BR = V_d, \end{aligned}$$

where $V_u = 5$ and $V_d = 0$ are the call's payoffs. Two equations, two unknowns.

Step 1. Find Δ. Subtract the second equation from the first. The cash terms BR cancel:

$$\Delta\,(S_0 u - S_0 d) = V_u - V_d, \qquad \text{so} \qquad \Delta = \frac{V_u - V_d}{S_0 u - S_0 d}.$$

For the call, $\Delta = \dfrac{5 - 0}{110 - 90} = \dfrac{5}{20} = 0.25$.

Read the formula. Δ is the change in the option's value divided by the change in the share's price: a slope. The share's two outcomes are 20 apart and the call's are 5 apart, so the copy needs a quarter of a share. This is the discrete version of the derivative $\partial V / \partial S$ (Lesson 2), and it will become the Black-Scholes delta.

Step 2. Find B. Put Δ back into the heads equation: $\Delta S_0 u + BR = V_u$, so

$$B = \frac{V_u - \Delta\, S_0 u}{R} = \frac{5 - 0.25 \times 110}{R} = \frac{5 - 27.5}{R} = \frac{-22.5}{R} = -22.2761.$$

B is negative: the copy **borrows** 22.2761 today and repays it with interest, $22.2761 \times R = 22.5$, in four months.

Step 3. Check both outcomes.

- After heads: $0.25 \times 110 - 22.5 = 27.5 - 22.5 = 5$. ✓
- After tails: $0.25 \times 90 - 22.5 = 22.5 - 22.5 = 0$. ✓

Step 4. The cost today. A quarter of a share costs 25, and the loan brings in 22.2761, so

$$V_0 = \Delta S_0 + B = 25 - 22.2761 = 2.7239.$$

Why the call must cost exactly this

Suppose the call traded at some other price C.

- **If $C > 2.7239$:** sell the call for C, and buy the copy for 2.7239. You pocket the difference today. In four months the copy pays exactly what you owe on the call, in both outcomes. That is an arbitrage.
- **If $C < 2.7239$:** do the opposite: buy the call, and *sell* the copy (short a quarter of a share, deposit 22.2761). Again you pocket the difference today, and the positions cancel later.

So the only price with no arbitrage is the cost of the copy. This is the **law of one price**: two things that pay the same in every world must cost the same.

Solving for the risk-neutral probability

The recipe above works, but it hides something. Put the formulas for Δ and B into $V_0 = \Delta S_0 + B$, for any payoffs V_u and V_d.

Step 1. Substitute.

$$V_0 = \frac{V_u - V_d}{S_0(u-d)}\,S_0 + \frac{1}{R}\Big(V_u - \frac{V_u - V_d}{S_0(u-d)}\,S_0 u\Big) = \frac{V_u - V_d}{u-d} + \frac{1}{R}\Big(V_u - \frac{(V_u - V_d)\,u}{u-d}\Big).$$

Step 2. Put everything over $R(u-d)$.

$$V_0 = \frac{1}{R(u-d)}\Big(R(V_u - V_d) + V_u(u-d) - (V_u - V_d)u\Big).$$

Step 3. Expand and collect V_u and V_d. Inside the bracket: $RV_u - RV_d + uV_u - dV_u - uV_u + uV_d$. The uV_u terms cancel, leaving

$$V_u(R-d) + V_d(u-R).$$

Step 4. Read off the weights.

$$V_0 = \frac{1}{R}\Big(\frac{R-d}{u-d}\,V_u + \frac{u-R}{u-d}\,V_d\Big).$$

Call the first weight $\tilde{p}$ ("p tilde"):

$$\boxed{\tilde{p} = \frac{R-d}{u-d}, \qquad 1-\tilde{p} = \frac{u-R}{u-d}, \qquad V_0 = \frac{\tilde{p}\,V_u + (1-\tilde{p})\,V_d}{R}}$$

(Check that the weights add up: $(R-d) + (u-R) = u-d$.)

The surprise. The price *is* a discounted average of the payoffs, just not with the real probabilities. It uses the weights $\tilde{p}$ and $1-\tilde{p}$, which come from u, d and R alone.

Numbers.

$$\tilde{p} = \frac{1.010050 - 0.9}{1.1 - 0.9} = \frac{0.110050}{0.2} = 0.550251, \qquad 1-\tilde{p} = 0.449749,$$

$$V_0 = \frac{0.550251 \times 5 + 0.449749 \times 0}{1.010050} = 2.7239,$$

the same as the cost of the copy.

Why the weights deserve to be called probabilities

They lie between 0 and 1. $\tilde{p} > 0$ exactly when $R > d$, and $\tilde{p} < 1$ exactly when $R < u$. So the no-arbitrage condition $d < R < u$ is precisely the condition that makes $\tilde{p}$ a genuine probability. (This is the one-step version of the **fundamental theorem of asset pricing**, Lesson 11: no arbitrage if and only if such a probability exists.)

Under them, the share earns the risk-free rate. Average the share's growth with the weights $\tilde{p}$:

$$\tilde{p}\,u + (1-\tilde{p})\,d = \tilde{p}(u-d) + d = (R-d) + d = R.$$

So $\big(\tilde{p}S_0u + (1-\tilde{p})S_0d\big)/R = S_0$: the share's price is also its discounted average future value, $\big(0.5503 \times 110 + 0.4497 \times 90\big)/R = 100.00$. In a world where investors didn't care about risk, every asset would be expected to grow like cash; $\tilde{p}$ is the probability that makes the share look like that. Hence the name **risk-neutral probability**.

Why "neutral" doesn't mean investors are neutral. Real investors expect the share to earn more than cash: under $p = 0.6$ its average growth is 1.02 per step, against $R = 1.0101$. The risk-neutral probability is not a claim about anyone's beliefs. It is a **pricing device**: the unique set of weights that turns "cost of the copy" into "discounted average".

The real probability never appears

Look back at the whole lesson: $p = 0.6$ was never used. Not in Δ, not in B, not in $\tilde{p}$. However likely heads is, the copy works in both outcomes and costs the same, so the call's price is the same.

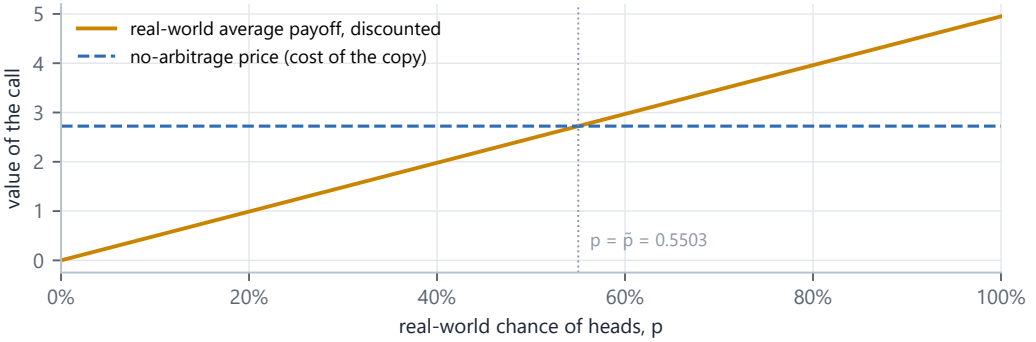


Figure 9.2. The call's no-arbitrage price (dashed) does not depend on the real chance of heads. The tempting answer, the real-world average payoff discounted, p × 5 / R (amber), depends on p and agrees with the price only at p = p̃.

At the real $p = 0.6$, the tempting answer is $0.6 \times 5/R = 2.9701$, which is 0.2463 too high. Anyone selling at that price would be beaten by someone who sells at 2.7239 and hedges.

Why intuition resists this. It feels as if a share that is more likely to rise should make a call more valuable. But think about who agrees on what. Two investors may disagree about p; they both agree that the share costs 100 today, because that is the market price. That price already contains everyone's views about p and about risk. The call is priced *relative to the share*: given the share's price, the copy's cost is fixed. Change the beliefs, and it is the share's price (or u and d) that would move, not the formula.

Trading against a wrong price

The notebook carries out the arbitrage for two wrong prices.

A dealer will buy the call at 3.00. Sell them the call (receive 3.00) and buy the copy (pay 2.7239: buy 0.25 shares for 25, borrow 22.2761). Profit today: 0.2761. In four months, after heads you owe 5 on the call and the copy is worth 5; after tails you owe nothing and the copy is worth nothing. Net: 0 in both cases.

A dealer will sell the call at 2.50. Buy it, and sell the copy: short 0.25 shares (receive 25) and deposit 22.2761. Profit today: 0.2239, and again the positions cancel at the end.

Trading against a wrong price.

```
def arbitrage(market_price):
    if market_price > rep["V0"]:      # sell the call, buy the copy
        return {"action": "sell call, buy copy", "today": market_price - rep["V0"],
                "end_H": rep["end_H"] - call_pay[0], "end_T": rep["end_T"] - call_pay[1]}
    return {"action": "buy call, sell copy", "today": rep["V0"] - market_price,
            "end_H": call_pay[0] - rep["end_H"], "end_T": call_pay[1] - rep["end_T"]}
```

The put, and a first look at put-call parity

The same recipe prices a **put** with strike 105, which pays $\max(105 - S_1, 0)$: 0 after heads and 15 after tails.

- $\Delta = (0 - 15)/(110 - 90) = -0.75$: the copy is **short** three quarters of a share, because the put gains when the share falls.
- $B = (0 - (-0.75)(110))/R = 82.5/R = 81.6791$: it holds cash.
- $P_0 = -0.75 \times 100 + 81.6791 = 6.6791$. With $\tilde{p}$: $(0.449749 \times 15)/R = 6.6791$. ✓

Now compare call and put. Holding a call and selling a put pays $\max(S_1 - 105, 0) - \max(105 - S_1, 0) = S_1 - 105$ in both outcomes. That is one share minus 105 owed in four months, which costs $S_0 - 105/R$ today. So

$$C - P = S_0 - \frac{105}{R}: \qquad 2.7239 - 6.6791 = -3.9552 = 100 - 103.9552.$$

Lesson 10 shows that this **put-call parity** holds on any tree, for any number of steps.

What to remember

- An **arbitrage** costs nothing, never loses and might win. Prices must not allow one.
- The share and cash alone are arbitrage-free only if $d < R < u$.
- **Copy** a payoff with Δ shares and B in cash: $\Delta = \dfrac{V_u - V_d}{S_0 u - S_0 d}$, $B = \dfrac{V_u - \Delta S_0 u}{R}$. Its price is $V_0 = \Delta S_0 + B$.
- Equivalently, $V_0 = \dfrac{\tilde{p} V_u + (1 - \tilde{p}) V_d}{R}$ with $\tilde{p} = \dfrac{R - d}{u - d}$.
- $\tilde{p} \in (0, 1)$ exactly when $d < R < u$; under $\tilde{p}$ the share grows on average like cash.
- The real probability p never enters the price.
- Teaching tree, one step: $\Delta = 0.25$, $\tilde{p} = 0.5503$, call 2.7239, put 6.6791.

Common mistakes

- **Pricing with the real probability.** pV_u/R is not a price unless p happens to equal $\tilde{p}$.

- **Forgetting to discount.** The average under $\tilde{p}$ is a value in four months; divide by R to bring it to today.
- **Thinking $\tilde{p}$ is a forecast.** It says nothing about how likely heads is. It is fixed by u, d and R.
- **Getting the sign of B wrong.** For a call, $B < 0$ (borrowing); for a put, $B > 0$. Check the copy in both outcomes.
- **Δ as a probability.** $\Delta = 0.25$ is a number of shares, the slope of the payoff, not a chance.
- **Ignoring the no-arbitrage condition.** If $R \geq u$, the formula gives $\tilde{p} \geq 1$ and the "price" means nothing: the market itself has an arbitrage.

Exercises

Solutions are in *Part 2: Solutions*.

9.1 Copy the put with strike 105 (it pays 0 after heads and 15 after tails): find Δ, B and the price. Why is Δ negative?

9.2 Suppose cash grew by 12% per step instead ($R = 1.12$) while the share still moves to 110 or 90. Write down an arbitrage and its profit in each outcome. Do the same for $R = 0.88$.

9.3 Solve the two copying equations for Δ and B in general, substitute into $V_0 = \Delta S_0 + B$, and show that $V_0 = \big(\tilde{p}V_u + (1-\tilde{p})V_d\big)/R$ with $\tilde{p} = (R-d)/(u-d)$.

9.4 Show that $\tilde{p}$ lies strictly between 0 and 1 exactly when $d < R < u$, and that $\big(\tilde{p}S_0 u + (1-\tilde{p})S_0 d\big)/R = S_0$.

9.5 A digital option pays 10 if the first toss is heads and nothing otherwise. Find its price and its Δ.

9.6 A dealer quotes the call at 3.10. Describe the arbitrage trade step by step, and find the profit today and the position after each outcome.

9.7 *(Code)* Write `one_period(S0, u, d, R, payoff)` that returns Δ, B and the price. Use it to check put-call parity $C - P = S_0 - K/R$ for strikes 92, 96, ..., 108.

9.8 *(Code)* Plot the discounted real-world average payoff pV_u/R and the no-arbitrage price as p runs from 0 to 1. Where do they cross?

Self-test

1. If the real chance of heads rose from 0.6 to 0.9, with the share still at 100 today, would the call's price change?
2. What does $\Delta = 0.25$ mean, in words?
3. What condition must u, d and R satisfy, and why?

Companion notebook: `s5e09_one_period_binomial.ipynb`. Every number in this chapter is computed by it.

LESSON 10 · VIDEO 10

The Multi-Period Binomial Model

Pricing a call that expires after three steps by working backward through the tree, one one-step problem at a time; the hedge at every node; why the backward values are the price; one formula for the whole tree; dynamic hedging along a path, and why it needs no new money; put-call parity; and many small steps, where the tree's price settles on the Black-Scholes value.

Why this lesson

Lesson 9 priced an option that lives for one step. Real options live for many steps, and the share can wander up and down before expiry. This lesson prices the course's call (strike 105, one year) on the full three-step teaching tree, and then on trees with hundreds of steps.

The method is **backward induction**: start at expiry, where the option's value is just its payoff, and step back one layer at a time, solving a one-step problem at every node. It is the tower rule of Lesson 8 with two changes: risk-neutral weights, and discounting.

Along the way we meet the idea behind every hedging desk: **dynamic hedging**. The copy of an option is not a fixed portfolio; it has to be adjusted after every move, and the adjustments pay for themselves. In Part 4 this becomes the Black-Scholes argument, with the steps shrunk to zero.

What you need from earlier lessons

- **The one-step formulas** (Lesson 9). For a node with share price S whose next values are V_u (after heads) and V_d (after tails):

$$\Delta = \frac{V_u - V_d}{Su - Sd}, \qquad V = \frac{\tilde{p}V_u + (1-\tilde{p})V_d}{R}, \qquad B = V - \Delta S,$$

with $\tilde{p} = (R-d)/(u-d) = 0.5503$ and $R = e^{0.01}$ on the teaching tree.

- **The tower rule** (Lesson 8). Averages can be taken one layer at a time.

- **Binomial coefficients.** $\binom{N}{k} = \frac{N!}{k!\,(N-k)!}$ counts the ways to place k heads among N tosses: $\binom{3}{0} = 1$, $\binom{3}{1} = 3$, $\binom{3}{2} = 3$, $\binom{3}{3} = 1$.
- **The normal distribution function** N (Lesson 4), for the Black-Scholes comparison at the end only.

The picture

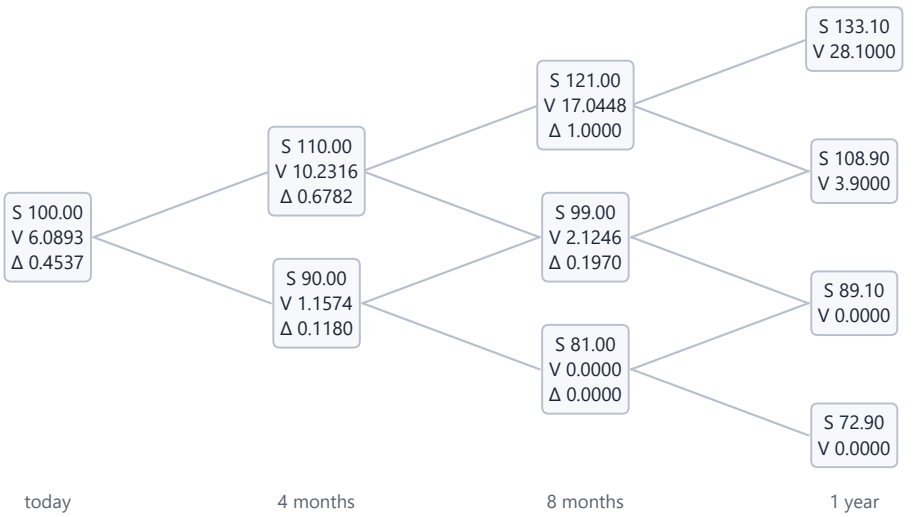


Figure 10.1. The three-step teaching tree for the call with strike 105. Each node shows the share price S, the call's value V and the number of shares Δ in its copy. Values are filled in from the right (expiry) to the left (today).

Read the tree from right to left. At one year the call's value is its payoff. At every earlier node it is the cost of copying the two values that can follow. The leftmost box is the price today: 6.0893.

Nodes

On the teaching tree, the price after n steps depends only on the number of heads k so far: $S_0 u^k d^{n-k}$. So a node is labelled by (n, k): step n, k heads. From node (n, k) the next node is $(n+1, k+1)$ after heads or $(n+1, k)$ after tails. HT and TH both lead to $(2, 1)$: the tree **recombines**, and three steps need only $1 + 2 + 3 + 4 = 10$ nodes.

We write $V_{n,k}$ for the option's value at node (n, k).

Working backward through three steps

Step 1. Expiry. At one year the call is worth its payoff, $\max(S_3 - 105, 0)$:

NODE	S_3	PAYOFF
(3, 3)	133.1	28.1
(3, 2)	108.9	3.9
(3, 1)	89.1	0.0
(3, 0)	72.9	0.0

Step 2. Eight months. Each time-2 node is a one-step problem whose two outcomes we now know.

Node (2, 2), price 121. The next values are 28.1 and 3.9:

$$V_{2,2} = \frac{0.5503 \times 28.1 + (1 - 0.5503) \times 3.9}{R} = 17.0448.$$

$$\Delta = \frac{28.1 - 3.9}{133.1 - 108.9} = \frac{24.2}{24.2} = 1.$$

$\Delta = 1$: the call is sure to be exercised from here, so it behaves exactly like owning the share and owing 105 in four months. Indeed $B = V - \Delta S =$ -103.9552 $= -105/R$.

Node (2, 1), price 99. The next values are 3.9 and 0:

$$V_{2,1} = \frac{0.5503 \times 3.9}{R} = 2.1246.$$

$$\Delta = \frac{3.9 - 0}{108.9 - 89.1} = \frac{3.9}{19.8} = 0.1970.$$

Node (2, 0), price 81. Both next values are 0, so $V_{2,0} = 0$ and $\Delta = 0$: the call is dead.

Step 3. Four months.

Node (1, 1), price 110, with next values 17.0448 and 2.1246:

$$V_{1,1} = \frac{0.5503 \times 17.0448 + (1 - 0.5503) \times 2.1246}{R} = 10.2316.$$

$$\Delta = \frac{17.0448 - 2.1246}{121 - 99} = 0.6782.$$

Node (1, 0), price 90, with next values 2.1246 and 0:

$$V_{1,0} = \frac{0.5503 \times 2.1246}{R} = 1.1574.$$

$$\Delta = \frac{2.1246}{99 - 81} = 0.1180.$$

Step 4. Today.

$$V_0 = \frac{0.5503 \times 10.2316 + (1 - 0.5503) \times 1.1574}{R} = 6.0893.$$

$$\Delta_0 = \frac{10.2316 - 1.1574}{110 - 90} = 0.4537.$$

The copy today holds 0.4537 shares and borrows the rest: B_0 = -39.2815.

Backward induction: value and copy at every node.

```
pt = (R - d) / (u - d)
S = lambda n, k_: S0_ * u**k_ * d ** (n - k_)


def backward(payoff, n_steps=steps):
    # Values V[(n, k)] and the copy (Delta, B) at each node before expiry.
    V = {(n_steps, k_): payoff(S(n_steps, k_)) for k_ in range(n_steps + 1)}
    hedge = {}
    for n in range(n_steps - 1, -1, -1):
        for k_ in range(n + 1):
            Vu_, Vd_ = V[(n + 1, k_ + 1)], V[(n + 1, k_)]
            V[(n, k_)] = (pt * Vu_ + (1 - pt) * Vd_) / R
            D_ = (Vu_ - Vd_) / (S(n + 1, k_ + 1) - S(n + 1, k_))
            hedge[(n, k_)] = {"Delta": D_, "B": V[(n, k_)] - D_ * S(n, k_)}
```

How Δ moves

Look at Δ down the tree. It is near 1 at the top (the call will almost surely be exercised), near 0 at the bottom (it almost surely won't), and in between in the middle. And it **changes** after every toss: from 0.4537 today to 0.6782 after heads or 0.1180 after tails. So the copy must be **rebalanced** at every step. That is dynamic hedging, below.

Why the backward values are the price

At each node the one-step argument says: *if* the option will be worth V_u and V_d one step later, it must be worth V now, or there is an arbitrage over that one step. At time 2 the values at time 3 are certain (they are the payoffs), so the time-2 values are forced. Given those, the time-1 values are forced, and so on back to today. This is a proof by **induction**, working backward.

The same argument gives a trading strategy: hold the node's copy, and at the next node switch to that node's copy. Two questions remain: does switching cost extra money, and does the strategy really end with the payoff? The section on hedging answers both.

One formula for the whole tree

Each backward step is a discounted $\tilde{p}$-average, so by the tower rule (Lesson 8) the whole thing is one discounted $\tilde{p}$-average over all paths:

$$V_0 = \frac{1}{R^3}\tilde{E}\big[\max(S_3 - 105, 0)\big].$$

Count the paths. A path with k heads has risk-neutral probability $\tilde{p}^k(1-\tilde{p})^{3-k}$, there are $\binom{3}{k}$ such paths, and they all end at $S_0 u^k d^{3-k}$. So for N steps,

$$V_0 = \frac{1}{R^N}\sum_{k=0}^{N}\binom{N}{k}\tilde{p}^k(1-\tilde{p})^{N-k}\max\left(S_0 u^k d^{N-k} - K, 0\right).$$

For the teaching tree:

The one-shot formula, term by term.

HEADS K	PATHS	RISK-NEUTRAL PROBABILITY	S_3	PAYOFF
3	1	0.1666	133.1	28.1
2	3	0.4085	108.9	3.9
1	3	0.3339	89.1	0.0
0	1	0.0910	72.9	0.0

$$V_0 = 0.970446 \times \big(0.1666 \times 28.1 + 0.4085 \times 3.9\big) = 6.0893,$$

the same as backward induction. The formula is quick for a European option, which depends only on the final price. Backward induction is more general: Lesson 12 uses it for American options, where the formula no longer applies.

Hedging along a path

Suppose you **sell** the call today for 6.0893 and want to be sure you can pay whatever it is worth in a year. Follow the copy:

1. **Today.** Buy $\Delta_0 = 0.4537$ shares. They cost 45.3709; you have 6.0893 from the sale, so you borrow the rest.
2. **After each toss.** Your cash (borrowed, so negative) grows by R. The share has moved. Look up the new node's Δ, and buy or sell shares to reach it, paying from (or into) the cash account.
3. **At expiry.** Sell everything and pay the call's holder.

Here is the path HTH, step by step. "Portfolio" is the value of your shares plus cash just before you rebalance; "option" is the backward-induction value at that node.

The hedge along HTH. At every step the portfolio is worth exactly the option.

STEP	TOSS	SHARE	Δ after	CASH AFTER	PORTFOLIO	OPTION
0		100.00	0.4537	-39.2815	6.0893	6.0893
1	H	110.00	0.6782	-64.3691	10.2316	10.2316
2	T	99.00	0.1970	-17.3754	2.1246	2.1246
3	H	108.90	0.0000	3.9000	3.9000	3.9000

At the end the portfolio is worth 3.9000, exactly the call's payoff, $108.9 - 105 = 3.9$. The notebook runs all eight paths: the largest error is 1.4e-14, rounding noise.

Why rebalancing is free

The hedge is **self-financing**: after the initial 6.0893, no money is added or taken out. Why doesn't buying more shares at a node cost anything extra?

At node $(1, 1)$, after heads, the old portfolio (Δ_0 shares and cash $B_0 R$) is worth

$$\Delta_0 \times 110 + B_0 R = V_{1,1},$$

because that is exactly what the copy was built to be worth after heads. The new portfolio costs $\Delta_{1,1} \times 110 + B_{1,1} = V_{1,1}$, by the definition of $B_{1,1}$. Same number: you sell the old portfolio and buy the new one with the proceeds. (Exercise 10.5 puts in the numbers.)

What the hedge does not depend on

The real probability p appears nowhere: not in $\tilde{p}$, not in any Δ. The hedge works in **every** world, whatever the coin's bias. That is the point of a hedge: the seller has no exposure to where the share goes.

Put-call parity

Claim. For a call and a put with the same strike K and the same expiry after N steps,

$$C - P = S_0 - \frac{K}{R^N}.$$

Why. At expiry, owning the call and having sold the put pays

$$\max(S_N - K, 0) - \max(K - S_N, 0) = S_N - K$$

in every world. (If $S_N > K$ the put pays 0 and the call $S_N - K$; if $S_N \le K$ the call pays 0 and the put $K - S_N$, which you owe.) Another way to get exactly $S_N - K$: own one share and owe K at expiry, which costs $S_0 - K/R^N$ today. Two payoffs equal in every world must cost the same (Lesson 9), so the claim follows.

Notice that the argument doesn't use the tree at all: parity holds in **any** arbitrage-free model.

Numbers. On the teaching tree the put, priced by backward induction, is worth 7.9861, so

$$C - P = 6.0893 - 7.9861 = -1.8968, \qquad S_0 - \frac{105}{R^3} = 100 - 101.8968 = -1.8968.$$

The deltas agree too: $\Delta_{\text{call}} - \Delta_{\text{put}} = 0.4537 - (-0.5463) = 1$, the delta of one share.

Many steps: the tree meets Black-Scholes

The teaching tree has huge steps: 10% every four months. A real pricing tree has hundreds of small steps, and its moves are set from the share's volatility.

The Cox-Ross-Rubinstein (CRR) tree. Split the year into N steps of length $\Delta t = T/N$ and set

$$u = e^{\sigma\sqrt{\Delta t}}, \qquad d = \frac{1}{u} = e^{-\sigma\sqrt{\Delta t}}, \qquad R = e^{r\Delta t}.$$

Why these moves. Each step changes the log price $\ln S$ by $+\sigma\sqrt{\Delta t}$ or $-\sigma\sqrt{\Delta t}$. With probabilities near one half, that step has variance about $\sigma^2\Delta t$, so over N steps the log price has variance about $N\sigma^2\Delta t = \sigma^2 T$: exactly the variance of the lognormal model of Lesson 5. And by the central limit theorem (Lesson 5), a sum of many small independent steps is close to normal. So as N grows, the tree's final price becomes lognormal, and its option prices approach the Black-Scholes formula (derived in Lesson 20):

$$C_{\text{BS}} = S_0 N(d_+) - K e^{-rT} N(d_-) = 7.1281.$$

With three steps, $u = e^{0.2/\sqrt{3}} = 1.1224$: a little bigger than the teaching tree's 1.1. The prices for more steps:

CRR call prices with the course's numbers, against Black-Scholes.

STEPS N	TREE PRICE	ERROR
3	7.5225	0.3944
10	7.2934	0.1653
30	7.1755	0.0474
100	7.1340	0.0059
300	7.1263	-0.0018
1000	7.1279	-0.0002

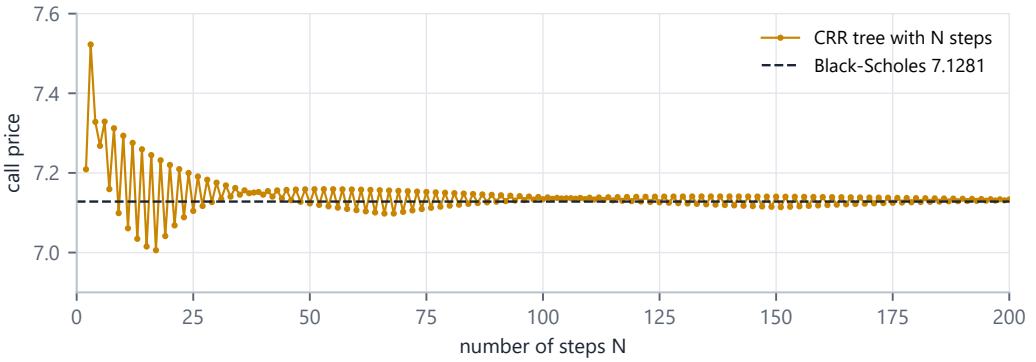


Figure 10.2. The CRR call price for N = 2 to 200 steps. It settles on the Black-Scholes value, but not smoothly: it zigzags between odd and even N and drifts in waves.

Why it zigzags. The tree's final prices sit on a grid, and the payoff has a kink at the strike. How the grid points fall around 105 changes with N, and for odd and even N the grid points are offset from each other. So the error swings between two "families". The swings shrink roughly in proportion to $1/N$, which is why banks use a few hundred steps, or smooth the payoff near the strike. Lesson 22 studies this convergence properly.

A CRR pricer in a dozen lines.

```
def crr(N, payoff=call, american=False):
    dt = T / N
    uu = math.exp(sigma * math.sqrt(dt)); dd = 1 / uu; RR = math.exp(r * dt)
    q = (RR - dd) / (uu - dd)
    s = S0 * uu ** np.arange(N, -1, -1) * dd ** np.arange(0, N + 1)
    v = np.array([payoff(x) for x in s])
    for n in range(N - 1, -1, -1):
        s = s[:-1] / uu
        v = (q * v[:-1] + (1 - q) * v[1:]) / RR
        if american:
            v = np.maximum(v, [payoff(x) for x in s])
    return float(v[0])
```

What to remember

- **Backward induction:** at expiry, value = payoff; at each earlier node, $V = (\tilde{p}V_u + (1-\tilde{p})V_d)/R$ and $\Delta = (V_u - V_d)/(Su - Sd)$.
- **One formula:** $V_0 = R^{-N}\sum_k \binom{N}{k}\tilde{p}^k(1-\tilde{p})^{N-k}\,\text{payoff}(S_0u^kd^{N-k})$, for options paid at expiry.
- **Dynamic hedging:** hold the node's Δ, rebalance after every step. It is self-financing and ends exactly at the payoff, in every world, whatever p is.
- **Put-call parity:** $C - P = S_0 - K/R^N$, in any arbitrage-free model.
- **CRR tree:** $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$; as $N \to \infty$ the price tends to Black-Scholes, zigzagging.
- Teaching tree: call 6.0893, $\Delta_0 = 0.4537$, put 7.9861.

Common mistakes

- **Discounting only once at the end of backward induction.** Each step back divides by R. (The one-shot formula divides by R^N once, because it does all the steps together.)
- **Keeping today's Δ all the way.** The copy has to follow the node's Δ. A hedge left alone is not a hedge: Exercise 10.8 shows how badly it can miss.
- **Using the real probability in backward induction.** That computes the average payoff (Lesson 8), not the price.
- **Thinking the hedge needs a view on the share.** It works in every world; its cost was fixed at the start.
- **Expecting the tree to converge smoothly.** It zigzags; compare odd with odd and even with even, or average two neighbouring N.

Exercises

Solutions are in *Part 2: Solutions*.

10.1 Price a call with strike 105 that expires after **two** steps (eight months), by backward induction on the teaching tree.

10.2 For the two-step call of Exercise 10.1, find Δ and the cash B at each of the three nodes before expiry. Why is Δ 0 at the lower node after one step?

10.3 Price the three-step put with strike 105 by backward induction and check put-call parity $C - P = S_0 - K/R^3$ with the call price from the lesson.

10.4 Write the one-shot formula for $N = 2$ and check that it gives your answer to Exercise 10.1.

10.5 In the hedge of the three-step call, show that rebalancing after the first toss (heads) costs nothing: the portfolio's value just before rebalancing equals the cost of the new portfolio.

10.6 A forward contract pays $S_3 - 105$ at the end. Show that its value at step n is $S_n - 105/R^{3-n}$ and that its Δ is 1 at every node. Why does a forward need no rebalancing?

10.7 *(Code)* Write a CRR pricer for N steps and tabulate the call price for $N = 50, 51, 52, \ldots, 60$. Describe how the error behaves for even and odd N.

10.8 *(Code)* Run the three-step hedge along all eight worlds and print the final error. Then repeat with a lazy hedge that keeps the first Δ to the end. How large do its errors get?

Self-test

1. Why do we work backward through the tree rather than forward?
2. Does the number of shares in the hedge depend on the real probability of heads?
3. A call and a put on the same share have the same strike and expiry. What is $C - P$?

Companion notebook: `s5e10_multi_period_binomial.ipynb`. Every number in this chapter is computed by it.

LESSON 11 · VIDEO 11

Martingales and Risk-Neutral Pricing

Martingales (fair games), submartingales and supermartingales; why the discounted share price and the discounted value of every option are martingales under the risk-neutral probability; the risk-neutral pricing formula; why no trading strategy can beat a martingale; the two fundamental theorems of asset pricing, and a market where the risk-neutral probability is not unique; the Markov property, and payoffs that depend on the whole path.

Why this lesson

Lessons 9 and 10 found prices by copying payoffs, node by node. This lesson steps back and asks what all those prices have in common. The answer is one word: **martingale**. Under the risk-neutral probability, every discounted price, of the share, of cash, of any option, is a fair game: its best forecast of tomorrow is today.

That one fact gives:

- the **risk-neutral pricing formula**, $V_n = \tilde{E}[V_N \mid \mathcal{F}_n]/R^{N-n}$, which is how Part 4 prices everything;
- a clean reason **why no arbitrage exists**: no strategy can turn a fair game into a sure profit;
- the **fundamental theorems of asset pricing**, which connect no arbitrage, risk-neutral probabilities and hedging.

The second half of the lesson looks at payoffs that depend on the whole path of the share, not just where it ends. They break the tree's neat recombining shape, and they show why the **Markov property** matters so much in practice.

What you need from earlier lessons

- **Filtrations and adapted processes** (Lesson 7). $\mathcal{F}_n$ is the information after n tosses; X_n is adapted if it is known at time n.

- **Conditional expectation and its rules** (Lesson 8): linearity, taking out what is known, independence, and the tower rule.
- **The one-step formulas** (Lesson 9). $\tilde{p} = (R - d)/(u - d) = 0.5503$, with $\tilde{p}u + (1 - \tilde{p})d = R$.
- **Backward induction and self-financing hedges** (Lesson 10).

We write $\tilde{E}$ for an expectation computed with $\tilde{p}$ in place of p, and $\tilde{E}[\,\cdot \mid \mathcal{F}_n]$ for the conditional version.

The picture

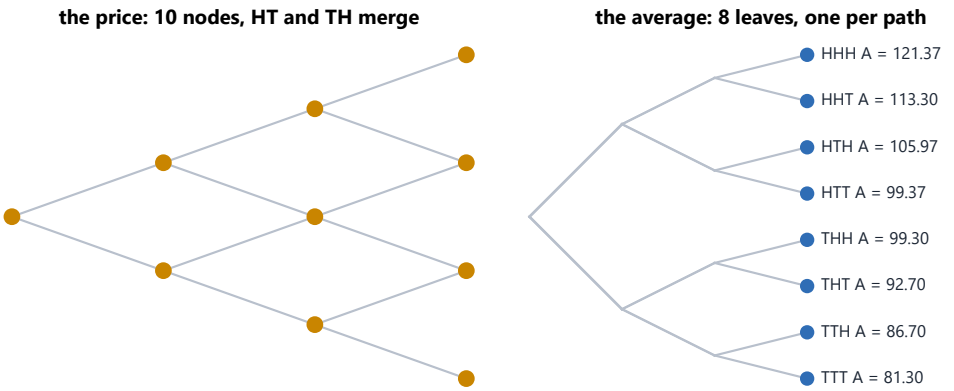


Figure 11.1. Left: the share price lives on a recombining tree, because HT and TH lead to the same price. Right: the average price along the path does not recombine; each of the eight paths ends with its own average, so a payoff on the average needs all eight leaves.

On the left, the future of the share depends only on where it is now, so paths that reach the same price can be merged. On the right, the quantity we care about remembers how the share got there. By the end of the lesson you will know which kind of tree a payoff needs, and why the left kind is such a gift.

Martingales

Definition. A process $M_0, M_1, \ldots, M_N$ is a **martingale** (with respect to a filtration $\mathcal{F}_n$ and a probability P) if

1. it is **adapted**: M_n is known at time n;
2. it has finite expectations, $E|M_n| < \infty$ (automatic on a finite tree);
3. **the best forecast of the next value is the current value:**

$$E[M_{n+1} \mid \mathcal{F}_n] = M_n \qquad \text{for every } n.$$

If instead $E[M_{n+1} \mid \mathcal{F}_n] \geq M_n$, it is a **submartingale** (it tends to rise); if $\leq$, a **supermartingale** (it tends to fall).

A fair game. Think of M_n as your wealth after n rounds of a game. A martingale is a game where, whatever has happened so far, your expected gain on the next round is zero. A casino game is a supermartingale for the player, and a submartingale for the casino. (The memory aid: a *super*martingale is *super* for the house.)

Two consequences.

Constant expectation. Take expectations of property 3 and use the tower rule: $E[M_{n+1}] = E\big[E[M_{n+1} \mid \mathcal{F}_n]\big] = E[M_n]$. So $E[M_n] = M_0$ for every n.

Forecasts any number of steps ahead. For $m > n$, apply property 3 repeatedly with the tower rule:

$$E[M_m \mid \mathcal{F}_n] = E\big[E[M_m \mid \mathcal{F}_{m-1}] \mid \mathcal{F}_n\big] = E[M_{m-1} \mid \mathcal{F}_n] = \cdots = M_n.$$

The best forecast of *any* future value is today's value.

Whether a process is a martingale depends on the probability. On the teaching tree, every process below satisfies $E[X_{n+1} \mid \mathcal{F}_n] = g\,X_n$ for a fixed factor g, so its type is read off from g:

The same tree, different processes and probabilities.

PROCESS	PROBABILITY	FORECAST FACTOR G	TYPE
S_n	p = 0.6	1.0200	submartingale
S_n	p = 0.5	1.0000	martingale
S_n	p̃	1.0101	submartingale
S_n / R^n	p = 0.6	1.0099	submartingale
S_n / R^n	p̃	1.0000	martingale
R^n / S_n	p̃	1.0100	submartingale

Read the rows:

- S_n **under the real** $p = 0.6$**:** the share is expected to grow 2% a step: a submartingale. It would be a martingale only with $p = 0.5$, since then $0.5(1.1) + 0.5(0.9) = 1$.
- S_n **under** $\tilde{p}$**:** it grows like cash, by R: still a submartingale.
- S_n/R^n **under** p**:** the share beats cash on average (1.02 against 1.0101): a submartingale. That extra growth is the reward investors demand for bearing risk.

- S_n/R^n **under $\tilde{p}$: a martingale.** This is the one that matters.
- R^n/S_n **under $\tilde{p}$:** the share measured in cash is a martingale, but cash measured in shares is not. (Part 4's change of numéraire, Lesson 27, picks a different probability to fix that.)

Discounted prices are martingales under p̃

Claim. Under $\tilde{p}$, the **discounted share price** $\tilde{S}_n = S_n/R^n$ is a martingale.

Proof. Write $S_{n+1} = S_n G_{n+1}$, where G_{n+1} is u or d according to toss $n+1$.

Step 1. S_n is known at time n, so take it out (Lesson 8, Rule 2):

$$\tilde{E}[S_{n+1} \mid \mathcal{F}_n] = S_n\, \tilde{E}[G_{n+1} \mid \mathcal{F}_n].$$

Step 2. Toss $n+1$ is independent of the first n tosses (Rule 3):

$$\tilde{E}[G_{n+1} \mid \mathcal{F}_n] = \tilde{E}[G_{n+1}] = \tilde{p}u + (1-\tilde{p})d = R.$$

Step 3. Divide by R^{n+1}:

$$\tilde{E}\Big[\frac{S_{n+1}}{R^{n+1}} \,\Big|\, \mathcal{F}_n\Big] = \frac{S_n R}{R^{n+1}} = \frac{S_n}{R^n}.$$

Claim. Under $\tilde{p}$, the **discounted value** V_n/R^n of any option priced by backward induction (Lesson 10) is a martingale.

Proof. Backward induction says $V_n = \big(\tilde{p}V_{n+1}^{\mathrm{H}} + (1-\tilde{p})V_{n+1}^{\mathrm{T}}\big)/R$ at every node, where V_{n+1}^{H} and V_{n+1}^{T} are the values at the two next nodes. The bracket is exactly $\tilde{E}[V_{n+1} \mid \mathcal{F}_n]$, so

$$V_n = \frac{\tilde{E}[V_{n+1} \mid \mathcal{F}_n]}{R}, \qquad \text{that is,} \qquad \frac{V_n}{R^n} = \tilde{E}\Big[\frac{V_{n+1}}{R^{n+1}} \,\Big|\, \mathcal{F}_n\Big].$$

Each step of backward induction is the martingale property. The notebook checks it at every node of the three-step call; the largest gap is 3.6e-15.

The pricing formula

Combine the two consequences of the martingale property with the fact that at expiry the option is worth its payoff V_N:

$$\frac{V_n}{R^n} = \tilde{E}\Big[\frac{V_N}{R^N} \,\Big|\, \mathcal{F}_n\Big], \qquad \text{so} \qquad \boxed{V_n = \frac{1}{R^{N-n}}\,\tilde{E}[V_N \mid \mathcal{F}_n]}$$

and in particular today, $V_0 = \tilde{E}[V_N]/R^N$.

The risk-neutral pricing formula: the price of a payoff is its risk-neutral average, discounted. Part 4 writes it as $V_t = e^{-r(T-t)}\tilde{E}[V_T \mid \mathcal{F}_t]$ and it is the backbone of every pricing method that follows: Monte Carlo averages it, trees and finite differences compute it backward, and the Black-Scholes formula is its closed form.

Path by path. For the three-step call, average the eight payoffs with the risk-neutral path probabilities:

The call's payoff in each world, with the risk-neutral and real probabilities.

WORLD	$\tilde{P}$	P	S_3	PAYOFF
HHH	0.1666	0.216	133.1	28.1
HHT	0.1362	0.144	108.9	3.9
HTH	0.1362	0.144	108.9	3.9
HTT	0.1113	0.096	89.1	0.0
THH	0.1362	0.144	108.9	3.9
THT	0.1113	0.096	89.1	0.0
TTH	0.1113	0.096	89.1	0.0
TTT	0.0910	0.064	72.9	0.0

$$V_0 = \frac{1}{R^3}\sum_{\omega}\tilde{P}(\omega)\max\big(S_3(\omega) - 105, 0\big) = 6.0893,$$

the price from Lesson 10. With the real probabilities instead, the discounted average would be 7.5252: not a price.

And the share itself? $\tilde{E}[S_3] = 103.0455 = 100R^3$: under $\tilde{p}$ the share is expected to grow exactly like cash.

No strategy beats a martingale

Claim. Let X_n be the value of any self-financing portfolio of shares and cash (Lesson 10): at each step you hold Δ_n shares, chosen with time-n information, and the rest in cash, and nothing is added or taken out. Then X_n/R^n is a martingale under $\tilde{p}$.

Why. From n to $n+1$ the portfolio becomes $X_{n+1} = \Delta_n S_{n+1} + (X_n - \Delta_n S_n)R$. Take $\tilde{E}[\,\cdot \mid \mathcal{F}_n]$: Δ_n and X_n are known, and $\tilde{E}[S_{n+1} \mid \mathcal{F}_n] = RS_n$, so

$$\tilde{E}[X_{n+1} \mid \mathcal{F}_n] = \Delta_n RS_n + (X_n - \Delta_n S_n)R = RX_n.$$

Divide by R^{n+1}.

So there is no arbitrage. Suppose a strategy starts at $X_0 = 0$ and never ends below 0. Its discounted value has constant expectation, so $\tilde{E}[X_N/R^N] = X_0 = 0$. An average of numbers that are never negative can only be 0 if every number with positive probability is 0. Since every world has positive probability under $\tilde{p}$, $X_N = 0$ in every world: no chance of a profit. That rules out arbitrage. (Exercise 11.6 writes this out.)

This is the deep reason the risk-neutral probability prices correctly: under it, trading is a fair game, and you cannot build something from nothing.

The fundamental theorems of asset pricing

The two directions just seen are the first of two theorems that summarise the whole theory.

First fundamental theorem. A market has no arbitrage **if and only if** there is a risk-neutral probability: a probability that gives every world positive weight, under which every discounted price is a martingale.

Second fundamental theorem. An arbitrage-free market is **complete** (every payoff can be copied by a self-financing strategy) **if and only if** the risk-neutral probability is **unique**.

The binomial tree is the friendliest case: $\tilde{p} = (R - d)/(u - d)$ is the only choice, every payoff can be copied, and every payoff has exactly one price.

When the risk-neutral probability is not unique

Give the one-step share a **third** outcome: it can go up 10%, stay at 100, or go down 10%. A risk-neutral probability (q_u, q_m, q_d) must make the weights add up to 1 and the share grow like cash:

$$q_u + q_m + q_d = 1, \qquad 1.1\,q_u + 1.0\,q_m + 0.9\,q_d = R.$$

Two equations, three unknowns. Subtracting the first from the second gives $0.1(q_u - q_d) = R - 1$, so $q_u - q_d = 0.1005$. Every q_u between 0.1005 and 0.5503 gives positive weights, and each is a legitimate risk-neutral probability.

The call with strike 105 pays 5 only if the share goes up, so its "price" $5q_u/R$ ranges from 0.4975 to 2.7239. Every price in that interval is free of arbitrage, and none is forced: with three outcomes and only two assets (share and cash), the call **cannot be copied**. This is an **incomplete market**. Stochastic volatility and jump models (Part 5) are incomplete in the same way, and pricing there needs extra information from the market.

Markov or path-dependent

Definition. A process X_n is **Markov** if its future depends on the past only through its present value: for every function f,

$$E[f(X_{n+1}) \mid \mathcal{F}_n] = E[f(X_{n+1}) \mid X_n].$$

The share is Markov. S_{n+1} is S_n times a factor that is independent of everything so far, so knowing the whole path adds nothing to knowing S_n. That is why the tree can merge HT and TH: from the price 99, the future looks the same either way. (Lesson 8 checked this for S_3.) In Part 4, the Markov property is what turns a pricing problem into a partial differential equation (Feynman-Kac, Lesson 28).

A payoff that remembers the path. An **Asian call** pays $\max(A - 105, 0)$ on the average price

$$A = \frac{S_1 + S_2 + S_3}{3}.$$

Paths HTH and THH both end at 108.9, but their averages are 105.9667 and 99.3000: the first pays, the second doesn't. So the value of an Asian call at node $(2, 1)$, price 99, depends on how you got there. The recombining tree cannot hold it.

Pricing it over all paths. The pricing formula still works; we just average over all $2^3 = 8$ paths:

Path-dependent payoffs on the teaching tree.

WORLD	$\tilde{P}$	AVERAGE A	ASIAN PAYOFF	HIGHEST M	LOOKBACK PAYOFF
HHH	0.1666	121.37	16.3667	133.1	28.1
HHT	0.1362	113.30	8.3000	121.0	16.0
HTH	0.1362	105.97	0.9667	110.0	5.0
HTT	0.1113	99.37	0.0000	110.0	5.0
THH	0.1362	99.30	0.0000	108.9	3.9
THT	0.1113	92.70	0.0000	100.0	0.0
TTH	0.1113	86.70	0.0000	100.0	0.0
TTT	0.0910	81.30	0.0000	100.0	0.0

$$\text{Asian call} = \frac{1}{R^3} \sum_{\omega} \tilde{P}(\omega) \max(A - 105, 0) = 3.8707.$$

A **lookback call**, paying $\max(M_3 - 105, 0)$ on the highest price $M_3 = \max(S_0, \ldots, S_3)$, is worth 8.3737 by the same method (Exercise 11.5).

The cost of remembering. Paths double with every step; nodes don't:

A recombining tree grows like N^2, the set of paths like 2^N.

STEPS	NODES IN THE RECOMBINING TREE	PATHS
3	10	8
10	66	1,024
20	231	1,048,576
50	1,326	1,125,899,906,842,624

With 50 steps, a path-by-path average is hopeless. Two ways out:

- **Enlarge the state.** The pair (price, running total) *is* Markov: from it you know everything the future payoff needs. Build the tree on pairs; it is bigger than the price tree but far smaller than the set of paths. (Similarly (price, maximum so far) for a lookback.)
- **Simulate.** Average the payoff over a large random sample of paths (Monte Carlo, Lessons 6 and 22).

What to remember

- A **martingale** is a fair game: $E[M_{n+1} \mid \mathcal{F}_n] = M_n$; then $E[M_m \mid \mathcal{F}_n] = M_n$ for all $m > n$, and $E[M_n] = M_0$. Sub- ($\geq$) and supermartingales ($\leq$) drift up and down.
- Whether a process is a martingale depends on the probability.
- Under $\tilde{p}$: the discounted share S_n/R^n, the discounted value of every option and the discounted value of every self-financing portfolio are martingales.
- **Pricing formula:** $V_n = \tilde{E}[V_N \mid \mathcal{F}_n]/R^{N-n}$; $V_0 = \tilde{E}[V_N]/R^N$.
- **FTAP 1:** no arbitrage $\iff$ a risk-neutral probability exists. **FTAP 2:** complete $\iff$ it is unique. With three outcomes and two assets, prices form an interval.
- The share is **Markov**, so its tree recombines. Path-dependent payoffs (Asian, lookback) need all paths, an enlarged Markov state, or simulation.

Common mistakes

- **"The share price is a martingale."** Not under the real probability, and not even under $\tilde{p}$ unless you discount it.
- **Forgetting the probability.** "Martingale" always means "martingale under some probability and filtration". Say which.

- **"A martingale doesn't move."** It moves all the time; it just has no predictable drift. S_n/R^n under $\tilde{p}$ is as random as the share.
- **Using the pricing formula with real probabilities.** The discounted real-world average is not a price: 7.5252, not 6.0893.
- **Pricing an Asian option on the recombining tree.** Merging HT and TH throws away the average. Check whether the payoff depends only on the final price.
- **"Incomplete means arbitrage."** An incomplete market can be free of arbitrage; it just has many consistent prices instead of one.

Exercises

Solutions are in *Part 2: Solutions*.

11.1 Under the real probability $p = 0.6$, show that S_n is a submartingale. For which p is S_n a martingale? Under $\tilde{p}$, is S_n itself (not discounted) a martingale?

11.2 Prove that S_n/R^n is a martingale under $\tilde{p}$ for any one-step factors with $d < R < u$, using *taking out what is known*.

11.3 Show that the backward-induction values satisfy $\tilde{E}[V_{n+1}/R^{n+1} \mid \mathcal{F}_n] = V_n/R^n$, and deduce $V_0 = \tilde{E}[V_3]/R^3$ with the tower rule.

11.4 Show that a martingale has constant expectation, $E[M_n] = M_0$, and that $E[M_m \mid \mathcal{F}_n] = M_n$ for every $m > n$, not just $m = n + 1$.

11.5 A lookback call pays $\max(M_3 - 105, 0)$, where M_3 is the highest of $S_0, \ldots, S_3$. Price it by hand over the eight paths under $\tilde{p}$.

11.6 Let X_n be the value of a self-financing portfolio of shares and cash. Show that X_n/R^n is a martingale under $\tilde{p}$, and use it to show that no such portfolio can start at 0, never end below 0, and end above 0 with positive probability.

11.7 *(Code)* Price the Asian call ($K = 105$, average of $S_1, \ldots, S_N$) on a CRR tree with the course's numbers by brute force over all 2^N paths, for $N = 4, 8, 12, 16$. How does the number of paths grow, and how many paths would $N = 252$ (daily steps) need?

11.8 *(Code)* On a 50-step CRR tree for the course's call, check the martingale property of the discounted value at every node, and print the largest gap.

Self-test

1. Under the real probability, is the discounted share price a martingale, a submartingale or a supermartingale?

2. Why do we discount before asking whether a price is a martingale?
3. Can an Asian call be priced on the recombining tree? Why or why not?

Companion notebook: `s5e11_martingales.ipynb`. Every number in this chapter is computed by it.

LESSON 12 · VIDEO 12

Change of Measure and American Options

Two probabilities on the same worlds and the ratio that connects them, the Radon-Nikodym derivative Z; why E[Z] = 1 and why $\tilde{P}$ averages are P averages weighted by Z; the process Z_n and equivalent measures; the state price density, which turns every price into a real-world average; American options, priced by comparing exercise with continuation at every node; why their discounted value is a supermartingale; and stopping times, and the best time to stop.

Why this lesson

Part 2 has used two probabilities on the same eight worlds: the real one, P (heads with probability 0.6), which describes how the share actually behaves, and the risk-neutral one, $\tilde{P}$ (heads with probability $\tilde{p}$), which prices. This lesson answers two questions that will matter a great deal in Part 4.

How are the two probabilities related? World by world, $\tilde{P}$ is P multiplied by a weight Z, the **Radon-Nikodym derivative**. In continuous time the same weight becomes an exponential of Brownian motion, and the statement that reweighting by it turns the real share into a risk-neutral one is **Girsanov's theorem** (Lesson 25). Here you can see every weight.

What if the holder can choose when to exercise? Most traded single-stock options are **American**: they can be exercised at any time before expiry. Pricing them means deciding, at every node, whether to exercise now or wait. The tree handles this with one small change to backward induction, and the theory behind it, **optimal stopping**, is a beautiful use of the martingale ideas of Lesson 11.

What you need from earlier lessons

- **Probability measures on the tree** (Lesson 7). A probability is fixed by world weights $P(\omega)$ adding to 1.

- **Conditional expectation and the tower rule** (Lesson 8).
- **The risk-neutral probability** (Lesson 9). $\tilde{p} = (R - d)/(u - d) = 0.5503$, so $\tilde{P}(\omega) = \tilde{p}^{\#\mathrm{H}}(1-\tilde{p})^{\#\mathrm{T}}$.
- **Backward induction** (Lesson 10) and **the pricing formula** (Lesson 11): $V_0 = \tilde{E}[V_N]/R^N$.
- **Martingales and supermartingales** (Lesson 11).

The picture

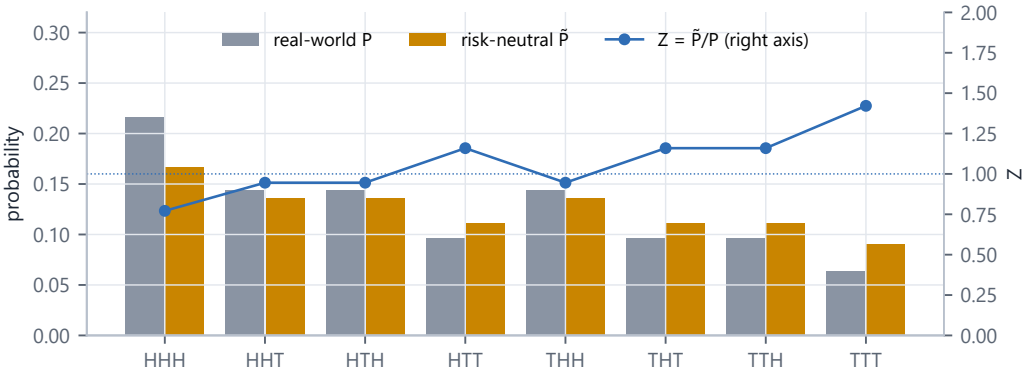


Figure 12.1. The eight worlds weighted two ways: by the real probability P (grey) and by the risk-neutral probability P̃ (amber). Their ratio Z = P̃/P (blue, right axis) is below 1 in worlds with many heads and above 1 in worlds with many tails.

Same worlds, different weights. The risk-neutral probability takes weight away from the good worlds (many heads) and gives it to the bad ones (many tails). The blue line, Z, says by how much. Everything in the first half of this lesson is about that line.

Reweighting the worlds: the Radon-Nikodym derivative

Definition. For two probabilities P and $\tilde{P}$ on the same finite set of worlds, with $P(\omega) > 0$ for every world, the **Radon-Nikodym derivative** of $\tilde{P}$ with respect to P is the random variable

$$Z(\omega) = \frac{\tilde{P}(\omega)}{P(\omega)}.$$

It is written $Z = \dfrac{d\tilde{P}}{dP}$, a notation that looks like a derivative because, in continuous settings, it plays the role of a ratio of densities.

Computing it on the tree. Both probabilities are products over the tosses, so their ratio is a product too:

$$Z(\omega) = \frac{\tilde{p}^{\#\mathrm{H}}(1-\tilde{p})^{\#\mathrm{T}}}{p^{\#\mathrm{H}}(1-p)^{\#\mathrm{T}}} = \left(\frac{\tilde{p}}{p}\right)^{\#\mathrm{H}}\left(\frac{1-\tilde{p}}{1-p}\right)^{\#\mathrm{T}}.$$

Each head multiplies the weight by $\tilde{p}/p = 0.9171$ (a little less), each tail by $(1-\tilde{p})/(1-p) = 1.1244$ (a little more).

The two probabilities and their ratio, world by world.

WORLD	S_3	P	$\tilde{P}$	Z
HHH	133.1	0.2160	0.1666	0.7713
HHT	108.9	0.1440	0.1362	0.9456
HTH	108.9	0.1440	0.1362	0.9456
HTT	89.1	0.0960	0.1113	1.1594
THH	108.9	0.1440	0.1362	0.9456
THT	89.1	0.0960	0.1113	1.1594
TTH	89.1	0.0960	0.1113	1.1594
TTT	72.9	0.0640	0.0910	1.4214

From $Z = 0.7713$ in HHH to 1.4214 in TTT.

Two properties.

1. $Z > 0$ in every world, since both probabilities are positive.
2. $E[Z] = 1$. Under P,

$$E[Z] = \sum_{\omega} Z(\omega)P(\omega) = \sum_{\omega} \frac{\tilde{P}(\omega)}{P(\omega)}P(\omega) = \sum_{\omega} \tilde{P}(\omega) = 1.$$

The notebook's value: 1.000000.

Conversely, *any* positive Z with $E[Z] = 1$ defines a new probability by $\tilde{P}(\omega) = Z(\omega)P(\omega)$: the weights are positive and add to 1. So changing the probability and choosing a positive random variable with average 1 are the same thing.

Averages under the new weights

Claim. For any random variable X,

$$\tilde{E}[X] = E[Z\,X].$$

Proof. $\tilde{E}[X] = \sum_\omega X(\omega)\tilde{P}(\omega) = \sum_\omega X(\omega)Z(\omega)P(\omega) = E[ZX]$.

So a risk-neutral average is a real-world average with Z as an extra weight. **Numbers:** for $X = S_3$, $\tilde{E}[S_3] = 103.0455$ and $E[ZS_3] = 103.0455$, while the unweighted real-world average is $E[S_3] = 106.1208$. Multiplying by Z pulls the average down, because Z is small exactly where S_3 is large.

Seen from time n: the process Z_n

At time n you have seen only the first n tosses. The part of Z you can know by then is its conditional expectation:

$$Z_n = E[Z \mid \mathcal{F}_n].$$

Claim. $Z_n = \left(\frac{\tilde{p}}{p}\right)^{\#\mathrm{H}} \left(\frac{1-\tilde{p}}{1-p}\right)^{\#\mathrm{T}}$, counted over the **first n tosses only**.

Why. Write Z as a product of one factor per toss: $Z = f_1 f_2 f_3$, where f_i is $\tilde{p}/p$ or $(1-\tilde{p})/(1-p)$ according to toss i. The first n factors are known at time n, so they come out (Lesson 8, Rule 2); the remaining ones depend only on later tosses, independent of $\mathcal{F}_n$, so their conditional expectation is their plain expectation (Rule 3). Each later factor has average

$$E[f_i] = p \cdot \frac{\tilde{p}}{p} + (1-p) \cdot \frac{1-\tilde{p}}{1-p} = \tilde{p} + (1-\tilde{p}) = 1.$$

So $Z_n = f_1 \cdots f_n \times 1$.

Z_n is a martingale under P. $Z_{n+1} = Z_n f_{n+1}$ with $E[f_{n+1} \mid \mathcal{F}_n] = 1$, so $E[Z_{n+1} \mid \mathcal{F}_n] = Z_n$. (The tower rule gives the same thing directly: $E[Z_{n+1} \mid \mathcal{F}_n] = E[E[Z \mid \mathcal{F}_{n+1}] \mid \mathcal{F}_n] = E[Z \mid \mathcal{F}_n]$.) It starts at $Z_0 = 1$ and ends at $Z_3 = Z$. Along HHH it goes 1, 0.9171, 0.8410, 0.7713; along TTT, 1, 1.1244, 1.2642, 1.4214.

In Lesson 25 this process becomes $Z_t = \exp\left(-\theta W_t - \frac{1}{2}\theta^2 t\right)$, the exponential martingale of Brownian motion, and its job is the same: to turn the real-world share into a risk-neutral one.

Equivalent probabilities

For $Z = \tilde{P}/P$ to make sense we needed $P(\omega) > 0$. More precisely, the two probabilities must agree on which events are **impossible**:

Definition. P and $\tilde{P}$ are **equivalent** if, for every event A, $P(A) = 0$ exactly when $\tilde{P}(A) = 0$.

They can disagree about *how likely* things are, but not about *whether they can happen*. This is essential for pricing. If $\tilde{P}$ gave probability 0 to a world that can really happen, a payoff of 1 in that world would get price 0: you could buy it for nothing and win with positive real probability, an arbitrage. That is why the first fundamental theorem (Lesson 11) asks for a risk-neutral probability with every world positive.

The Radon-Nikodym theorem (stated, not proved here) says that for equivalent probabilities on any probability space, finite or not, there is a positive random variable Z with $E[Z] = 1$ such that $\tilde{P}(A) = E[Z\mathbf{1}_A]$ for every event A. On the tree, Z is the ratio of world weights. In continuous time, where single paths have probability 0 and ratios of world weights are meaningless, the theorem still delivers Z.

The state price density

Combine the reweighting with discounting. Put

$$\zeta(\omega) = \frac{Z(\omega)}{R^N} \qquad (\zeta \text{ is the Greek letter zeta}).$$

Then the pricing formula becomes a **real-world** average:

$$V_0 = \frac{\tilde{E}[V_N]}{R^N} = \frac{E[Z\,V_N]}{R^N} = E[\zeta\,V_N].$$

What ζ means. Take a security that pays 1 in world ω and nothing elsewhere (an **Arrow-Debreu security**). Its price is $\tilde{P}(\omega)/R^N$, the **state price** of ω. Dividing by the real probability gives $\zeta(\omega)$: the price of a dollar in world ω, *per unit of real probability*. That's why ζ is called the **state price density** (or **stochastic discount factor**).

Numbers. ζ runs from 0.7485 in HHH to 1.3794 in TTT. The state prices add up to 0.970446, which is $1/R^3$: buying a dollar in every world is the same as buying a risk-free dollar. And for the call,

$$E[\zeta\,\max(S_3 - 105, 0)] = 6.0893,$$

the price from Lesson 10.

Why ζ is largest in the worst world. A dollar that arrives when the share has crashed (TTT) is worth more than a dollar that arrives when everyone is rich (HHH): it arrives when it is most needed. Investors pay up for protection against bad worlds and demand a discount for exposure to them. ζ records that preference, world by world. Under $\tilde{P}$ it is folded into the probabilities, which is why risk-neutral pricing needs no explicit attitude to risk.

American options

Definition. A **European** option can be exercised only at expiry. An **American** option can be exercised at any step up to expiry; the holder chooses when.

A put is the natural example. If the share has fallen to 81, a put with strike 105 pays 24 if exercised now. Waiting risks the share recovering. Exercising now also gets you the 24 earlier, to earn interest.

Backward induction with a choice

At each node the holder compares two numbers:

- the **intrinsic value**, what exercising pays now: $\max(105 - S_n, 0)$;
- the **continuation value**, what keeping the option is worth: the discounted risk-neutral average of the next step's values, exactly as in Lesson 10.

A rational holder takes the larger:

$$V_n = \max\left(\underbrace{105 - S_n}_{\text{exercise}}, \underbrace{\frac{\tilde{p}\, V_{n+1}^{\mathrm{H}} + (1-\tilde{p})\, V_{n+1}^{\mathrm{T}}}{R}}_{\text{continue}} \right).$$

At expiry there is no continuation, so V_N is the payoff.

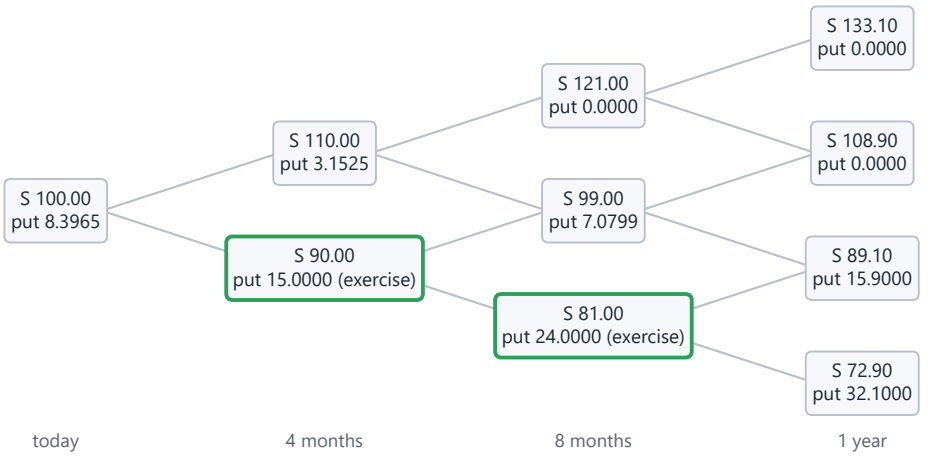


Figure 12.2. The American put with strike 105 on the teaching tree. Each node shows the share price and the put's value. At the framed nodes, exercising beats waiting.

Worked example: the three-step American put

Step 1. Expiry. The put pays $105 - S_3$ when positive: 0.0, 0.0, 15.9 and 32.1 at the prices 133.1, 108.9, 89.1 and 72.9.

Step 2. Eight months.

- *Node 121.* Both next values are 0: continuation 0, intrinsic 0. Value 0.
- *Node 99.* Continuation $(0.5503 \times 0 + (1 - 0.5503) \times 15.9)/R = 7.0799$. Intrinsic $105 - 99 = 6$. **Wait**: value 7.0799.
- *Node 81.* Continuation $(0.5503 \times 15.9 + (1 - 0.5503) \times 32.1)/R = 22.9552$. Intrinsic $105 - 81 = 24$. **Exercise**: value 24.

Step 3. Four months.

- *Node 110.* Continuation $(0.5503 \times 0 + (1 - 0.5503) \times 7.0799)/R = 3.1525$; intrinsic 0. Wait: 3.1525.
- *Node 90.* Continuation $(0.5503 \times 7.0799 + (1 - 0.5503) \times 24)/R = 14.5435$; intrinsic $105 - 90 = 15$. **Exercise**: value 15.

Step 4. Today. Continuation $(0.5503 \times 3.1525 + (1 - 0.5503) \times 15)/R = 8.3965$; intrinsic 5. Wait. **The American put is worth 8.3965.**

The European put (same tree, no early exercise) is worth 7.9861. The right to exercise early is worth the difference, the **early-exercise premium**: 0.4104.

Why exercise at 90? By exercising you get 15 now. By waiting you get a position worth 14.5435: you keep some chance of an even bigger payoff, but you give up four months' interest on 105 and you risk the share recovering. Deep in the money, the interest wins.

The American call: never exercised early

For a call on a share that pays no dividends, early exercise never helps. At any node the continuation value is at least what a forward would give, $S_n - 105/R^{N-n}$, which is more than the intrinsic value $S_n - 105$ because $105/R^{N-n} < 105$. So waiting always beats exercising, and the American call is worth the same as the European call. (Exercise 12.5 writes out the argument.) Intuitively: exercising a call means paying the strike early, and paying early only loses interest.

Supermartingales and the best time to stop

The discounted American value is a supermartingale

At every node, $V_n \geq$ continuation value $= \tilde{E}[V_{n+1} \mid \mathcal{F}_n]/R$. Dividing by R^n:

$$\tilde{E}\left[\frac{V_{n+1}}{R^{n+1}} \,\middle|\, \mathcal{F}_n\right] \leq \frac{V_n}{R^n}.$$

So **the discounted American value is a supermartingale** under $\tilde{p}$. It drifts down exactly at the nodes where exercise is better than waiting: continuation minus value is $22.9552 - 24 =$ -1.0448 at node 81, and $14.5435 - 15$ at node 90. At every other node the drift is 0.

Stopping times

Definition. A **stopping time** is a random time τ (a step number, depending on the world) such that the decision "stop at step n" can be made with time-n information: $\{\tau = n\} \in \mathcal{F}_n$ for every n.

"Exercise the first time the share falls below 95" is a stopping time. "Exercise at the step where the share is lowest" is not: you only know which step that was after the fact.

The best stopping time

The holder of an American option chooses a stopping time τ and receives the intrinsic value at τ. Its value today, discounted, is $\tilde{E}\left[\max(105 - S_\tau, 0)/R^\tau\right]$. Two facts hold:

1. **No rule beats the backward-induction price.** For any stopping time, $\tilde{E}[\text{payoff at } \tau/R^\tau] \le V_0$. (*Why:* the payoff never exceeds V_τ, and a supermartingale stopped at any stopping time still has expectation at most where it started.)
2. **One rule achieves it:** stop the first time the value equals the intrinsic value, $\tau^* = \min\{n : V_n = 105 - S_n\}$. Before τ^*, every node is a "continue" node, where the discounted value is a martingale; so stopped at τ^*, it is a martingale, and its expectation stays at V_0.

So the American price is the value of the **best stopping rule**, and the tree finds that rule as a by-product: the framed nodes in the figure.

Comparing rules. The notebook computes the risk-neutral average discounted payoff of four rules:

RULE	VALUE
Stop at the framed nodes (τ^*)	8.3965
Exercise as soon as the put is in the money	8.1346
Wait until expiry (the European put)	7.9861
Exercise immediately	5.0000

Exercising greedily, at 99 after HT, gives up value: there the put is worth 7.0799 alive but only 6 dead.

Many steps

With the course's numbers on a CRR tree (Lesson 10), the American put with strike 105 settles as the number of steps grows:

American and European puts, strike 105, one year, σ = 20%, r = 3%.

STEPS N	AMERICAN PUT	EUROPEAN PUT	EARLY-EXERCISE PREMIUM
10	9.6466	9.1901	0.4564
50	9.4760	9.0202	0.4558
100	9.4875	9.0308	0.4567
500	9.4795	9.0269	0.4525
1000	9.4776	9.0247	0.4529

The European column approaches the Black-Scholes put, 9.0248. The American put has no closed-form formula; the tree (or the finite-difference grids of Lesson 29) is how it is priced in practice. Its premium, about 0.45, is the value of being able to choose when to stop.

What to remember

- Two probabilities on the same worlds are linked by $Z = \tilde{P}/P$, the **Radon-Nikodym derivative**: $Z > 0$, $E[Z] = 1$, and $\tilde{E}[X] = E[ZX]$.
- On the tree, $Z = (\tilde{p}/p)^{\#\mathrm{H}}((1-\tilde{p})/(1-p))^{\#\mathrm{T}}$; $Z_n = E[Z \mid \mathcal{F}_n]$ uses the first n tosses and is a P-martingale.
- Pricing needs **equivalent** probabilities: same impossible events.
- **State price density** $\zeta = Z/R^N$: $V_0 = E[\zeta V_N]$, a real-world average; ζ is largest in the worst worlds.
- **American options:** $V_n = \max(\text{intrinsic}, \text{continuation})$ at each node. The discounted value is a **supermartingale** under $\tilde{p}$.
- The best exercise rule stops the first time value = intrinsic; no **stopping time** does better.
- Without dividends, an American call is never exercised early. An American put can be.
- Teaching tree: American put 8.3965, European 7.9861; exercise at 90 and 81.

Common mistakes

- **Writing** $Z = P/\tilde{P}$**.** $Z = d\tilde{P}/dP$ is new over old: $\tilde{E}[X] = E[ZX]$.

- **Forgetting $E[Z] = 1$.** A weight that doesn't average to 1 does not define a probability.
- **Reweighting with a Z that is zero somewhere real.** The new probability would call possible worlds impossible; pricing breaks.
- **Comparing exercise with the European value.** The continuation value at a node uses the *American* values at the next step, not the European ones.
- **Exercising as soon as the put is in the money.** It is often better to wait; the tree tells you where.
- **"An American option is a European option plus a bit."** The premium is real, but it comes from a different calculation: you cannot get the American price from the European formula.

Exercises

Solutions are in *Part 2: Solutions*.

12.1 Compute $Z = \tilde{P}/P$ for the eight worlds by hand (write it as $(\tilde{p}/p)^{\#\mathrm{H}}((1-\tilde{p})/(1-p))^{\#\mathrm{T}}$) and check that $E[Z] = 1$.

12.2 Prove that $\tilde{E}[X] = E[ZX]$ for any random variable X on the tree, and check it for $X = S_3$.

12.3 Show that $Z_n = E[Z \mid \mathcal{F}_n]$ equals $(\tilde{p}/p)^{\#\mathrm{H}}((1-\tilde{p})/(1-p))^{\#\mathrm{T}}$ over the first n tosses, and that Z_n is a martingale under P.

12.4 Two probabilities are **equivalent** if they agree on which events have probability 0. Why must $\tilde{P}$ be equivalent to P for pricing to make sense? What goes wrong if the real probability of heads is $p = 1$?

12.5 Show that an American call on this share (no dividends) is never exercised early: at every node its continuation value is at least $S_n - 105/R^{3-n} > S_n - 105$. So its price equals the European call's.

12.6 Price the American put with strike 105 that expires after **two** steps by hand. At which node is it exercised early?

12.7 *(Code)* With the course's numbers, compute the American and European puts for $N = 100, 200, \ldots, 1000$, and find, at the step closest to six months, the highest share price at which exercising is best (the **exercise boundary**).

12.8 *(Code)* Simulate 200,000 paths of the three-step teaching tree under $\tilde{p}$ and estimate the average discounted payoff of the best stopping rule and of *exercise as soon as in the money*. Compare with the exact values.

Self-test

1. What is $E[Z]$, and why?
2. For the American put on the teaching tree, where is early exercise optimal, and what is it worth?
3. Under $\tilde{p}$, is the discounted American put value a martingale, a submartingale or a supermartingale?

Companion notebook: `s5e12_change_of_measure_american.ipynb`. Every number in this chapter is computed by it.

Formula sheet and glossary

Formula sheet

Information (Lesson 7)

- A partition into b blocks generates a σ-algebra of 2^b events; after n tosses there are 2^n blocks.
- σ-algebra: contains Ω; closed under complements and countable unions. Filtration: $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \cdots$.
- X is $\mathcal{F}_n$-measurable $\iff$ it takes one value on each time-n block.
- Probability: $P(A) \geq 0$, $P(\Omega) = 1$, additive over disjoint events; $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

Conditional expectation (Lesson 8)

- On a block B: $E[X \mid \mathcal{F}_n] = \sum_{\omega \in B} X(\omega) P(\omega)/P(B)$.
- Definition: $E[X \mid \mathcal{G}]$ is $\mathcal{G}$-measurable and $E\big[E[X \mid \mathcal{G}]\mathbf{1}_A\big] = E[X\mathbf{1}_A]$ for all $A \in \mathcal{G}$.
- Linearity; $E[YX \mid \mathcal{G}] = YE[X \mid \mathcal{G}]$ for known Y; $E[X \mid \mathcal{G}] = E[X]$ for independent X.
- Tower: $E\big[E[X \mid \mathcal{G}] \mid \mathcal{H}\big] = E[X \mid \mathcal{H}]$ for $\mathcal{H} \subseteq \mathcal{G}$; $E\big[E[X \mid \mathcal{G}]\big] = E[X]$.
- On the tree: $E[S_{n+1} \mid \mathcal{F}_n] = \big(pu + (1-p)d\big) S_n$.

One step (Lesson 9)

- No arbitrage: $d < R < u$.
- $\Delta = \dfrac{V_u - V_d}{S_0 u - S_0 d}$, $B = \dfrac{V_u - \Delta S_0 u}{R}$, $V_0 = \Delta S_0 + B$.
- $\tilde{p} = \dfrac{R - d}{u - d}$, $V_0 = \dfrac{\tilde{p} V_u + (1-\tilde{p}) V_d}{R}$, $\tilde{p}u + (1-\tilde{p})d = R$.

Many steps (Lesson 10)

- Backward induction: $V_{n,k} = \big(\tilde{p} V_{n+1,k+1} + (1-\tilde{p}) V_{n+1,k}\big)/R$.
- One shot: $V_0 = R^{-N} \sum_k \binom{N}{k} \tilde{p}^{\,k} (1-\tilde{p})^{N-k} \,\text{payoff}(S_0 u^k d^{\,N-k})$.
- Self-financing: $X_{n+1} = \Delta_n S_{n+1} + (X_n - \Delta_n S_n) R$.
- Put-call parity: $C - P = S_0 - K/R^N$.
- CRR tree: $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$, $R = e^{r\Delta t}$.

Martingales (Lesson 11)

- Martingale: $E[M_{n+1} \mid \mathcal{F}_n] = M_n$; then $E[M_m \mid \mathcal{F}_n] = M_n$ $(m > n)$ and $E[M_n] = M_0$.
- Under $\tilde{p}$: S_n/R^n, V_n/R^n and every self-financing X_n/R^n are martingales.
- Pricing: $V_n = \tilde{E}[V_N \mid \mathcal{F}_n]/R^{N-n}$.
- FTAP: no arbitrage $\iff$ a risk-neutral probability exists; complete $\iff$ it is unique.

Change of measure and American options (Lesson 12)

- $Z = d\tilde{P}/dP = \tilde{P}(\omega)/P(\omega)$; $Z > 0$, $E[Z] = 1$, $\tilde{E}[X] = E[ZX]$.
- Tree: $Z_n = (\tilde{p}/p)^{\#\mathrm{H}}\big((1-\tilde{p})/(1-p)\big)^{\#\mathrm{T}}$ over the first n tosses; a P-martingale.
- State price density $\zeta = Z/R^N$: $V_0 = E[\zeta V_N]$.
- American: $V_n = \max\big(\text{intrinsic},\ (\tilde{p}V^{\mathrm{H}}_{n+1} + (1-\tilde{p})V^{\mathrm{T}}_{n+1})/R\big)$; V_n/R^n is a $\tilde{P}$-supermartingale; optimal $\tau^* = \min\{n : V_n = \text{intrinsic}\}$.

The teaching tree

Share price today, S_0	100
Up and down factors, u and d	1.1 and 0.9
Steps	3, of four months each
Real-world chance of heads, p	0.6
Cash growth per step, $R = e^{0.03/3}$	1.010050
Risk-neutral chance of heads, $\tilde{p}$	0.550251
Strike, K	105
European call, put	6.0893, 7.9861
American put	8.3965

Glossary

Adapted process. A process whose value at time n is known at time n.

American option. An option that can be exercised at any step up to expiry.

Arbitrage. A trade that costs nothing, never loses and might win.

Arrow-Debreu security. A security that pays 1 in one world and 0 in all others; its price is that world's state price.

Backward induction. Valuing an option from expiry back to today, one step at a time.

Complete market. One in which every payoff can be copied by a self-financing strategy.

Conditional expectation. $E[X \mid \mathcal{G}]$: the average of X given the information $\mathcal{G}$; on a tree, the average over the block you are in.

Continuation value. The value of keeping an American option alive one more step: the discounted risk-neutral average of the next values.

Copy (replicating portfolio). A portfolio of shares and cash with the same payoff as an option in every world.

Delta (Δ). The number of shares in the copy: change in option value over change in share price.

Equivalent probabilities. Probabilities with the same events of probability 0.

Event. A set of worlds.

Filtration. A growing sequence of σ-algebras, $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \cdots$: information over time.

Fundamental theorems of asset pricing. No arbitrage $\iff$ a risk-neutral probability exists; completeness $\iff$ it is unique.

Incomplete market. One where some payoffs cannot be copied; risk-neutral probabilities, and prices, are not unique.

Intrinsic value. What exercising an option pays right now.

Markov process. One whose future depends on the past only through its present value.

Martingale. A process whose best forecast of any future value is its current value. Submartingale: forecasts at least the current value; supermartingale: at most.

Measurable. X is $\mathcal{F}$-measurable if its value can be computed from the information $\mathcal{F}$.

Partition. A split of Ω into blocks that don't overlap and cover everything.

Path-dependent payoff. One that depends on the whole price path, not just the final price (Asian, lookback).

Probability measure. A function on events that is non-negative, gives Ω probability 1 and adds over disjoint events.

Put-call parity. $C - P = S_0 - K/R^N$ for a call and a put with the same strike and expiry.

Radon-Nikodym derivative. $Z = d\tilde{P}/dP$, the weight that turns one probability into another: $\tilde{E}[X] = E[ZX]$.

Recombining tree. A tree in which an up move followed by a down move reaches the same node as a down move followed by an up move.

Risk-neutral probability. The probability under which every discounted price is a martingale; on the tree, $\tilde{p} = (R - d)/(u - d)$.

Self-financing strategy. A trading strategy with no money added or taken out after the start.

σ-algebra. A collection of events containing Ω, closed under complements and countable unions: a description of information.

State price density. $\zeta = Z/R^N$; prices are real-world averages weighted by ζ.

Stopping time. A random time τ with $\{\tau = n\}$ known at time n.

World (outcome). One complete sequence of tosses; Ω is the set of all worlds.

Further reading

- Steven E. Shreve, *Stochastic Calculus for Finance I: The Binomial Asset Pricing Model*, Chapters 1 to 4: the same ground in more depth.
- Martin Baxter and Andrew Rennie, *Financial Calculus*, Chapter 2: a short, lively tour of discrete processes and martingales.
- David Williams, *Probability with Martingales*: for readers who want the measure theory behind Lesson 7 done properly.